

Quantum Mechanics is Deterministic

The Geometric Reading of Quantum Mechanics

*Fundamental Fractal Geometric
Field Theory (FFGFT)*

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<https://github.com/jpascher/T0-Time-Mass-Duality>

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Note: This book presents the state of the FFGFT corpus (September 2026). The underlying documents (Doc. 020–365) are freely available in the repository. Status markers: [K] confirmed · [B] proven · [S] speculative · [E] established physics · [X] tested negative.

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Part I

Foundations

Chapter 1

The Geometric Foundation

The fundamental relation of the FFGFT is

$$\tilde{T} \cdot m = 1, \quad (1.1)$$

where $\tilde{T}$ is a compact time coordinate and m is the mass of a mode. [SETZUNG] This relation is not a derived equation, but the geometric postulate from which everything else follows.

1.1 The Carrier: $T^4/\mathbb{Z}_3$

The FFGFT operates on a compact torus $T^4 = (S^1)^4$ with a $\mathbb{Z}_3$ lattice symmetry. This carrier has two essential properties.

Compactness. Each coordinate is periodic. This has an immediate consequence: The admissible modes are discrete — not because nature is postulated to be discrete, but because periodicity forces discrete Fourier modes. The Fourier modes of the carrier yield exactly the particle spectra of the Standard Model. [K]

$\mathbb{Z}_3$ symmetry. The quotienting by $\mathbb{Z}_3$ creates nine fixed points and a triality symmetry with $|\text{Aut}(D_4)| = 1152 = 2^7 \cdot 3^2$. This symmetry becomes visible in the spectral operator and controls which modes are stable. [B]

1.2 The Structure Parameter ξ

The only free parameter of the theory is

$$\xi = \frac{4}{30000} = \frac{1}{7500} \approx 1,333 \times 10^{-4}. \quad (1.2)$$

It is not a calibration quantity, but follows from the Galois group orders of the carrier: Specifically, $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4$. [K] (Doc. 338, Doc. 353)

All masses, couplings, and corrections of the FFGFT can be derived from ξ ; SI anchor values (electron mass, H_0) serve only for conversion into SI units, not for derivation.

1.3 The Relation $\tilde{T} \cdot m = 1$ and Its Consequences

Equation (1.1) has three immediate consequences.

Mode-bound time scale. Each mode with mass m_i carries its own compact time scale $\tilde{T}_i = 1/m_i$. [B] Time is not an external parameter, but a mode-specific property of the carrier. The Pauli theorem and the missing time operator of standard QM are not a problem in this reading: On a compact carrier, the Pauli argument does not apply (Doc. 334).

Mass arithmetic from geometry. The ratio $(m_\mu/m_e)^2 = 43200$ follows both geometrically from the winding numbers and from the Galois group orders $|GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)|$. [K] (Doc. 338)

Fractal correction. The transition from the compact carrier to the SI scale brings a recursive correction $K_{\text{frak}} = 1 - 100\xi \approx 0,9867$, which follows from the Type III pullback structure of the carrier. [B] (Doc. 333)

***For all readers:** Imagine physics does not happen on an infinitely long straight line, but on a drum — a surface that is closed in itself. On a drum, only certain vibrations are possible: those that return to where they started after one revolution. Exactly like this T^4 works: Only modes that "fit" on the torus exist — and that is exactly the particles that we measure.*

Sources: Doc. 230, 307, 314, 338, 353, 365

Chapter 2

States, Measurement, and Time

2.1 States as Mode Configurations

In standard QM, a state is a vector $|\psi\rangle$ in a Hilbert space $\mathcal{H}$. In the FFGFT, a state is a mode configuration (z, r, θ) on the carrier $T^4/\mathbb{Z}_3$. The connection is bijective: The spectral operator

$$\hat{F} = \sum_k f_k |k\rangle\langle k|, \quad (2.1)$$

with modes k from the lattice D_4 is self-adjoint on $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$. [B] (Doc. 327) Every QM calculation can be mapped into FFGFT language and vice versa; the formalisms yield the same statistical predictions. [B]

The Translation Table

Hilbert Space QM	FFGFT
Qubit $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$	Mode $(z, r, \theta) \in S^1 \times \mathbb{R}_{>0} \times S^1$
Bloch sphere S^2	Cylindrical fiber (r, θ)
Superposition	Intermediate torsion of a mode
Tensor product $ \psi\rangle \otimes \phi\rangle$	Independent mode configurations
Entanglement	Topological connection (non-reducible)
Unitary evolution	Phase rotation on T^4
Measurement / collapse	Pole transition to $z = \pm 1$
Born rule $P = \alpha ^2$	Statistical average over unknown phase θ

2.2 Measurement Without the Collapse Postulate

The collapse of the wave function is a postulate in standard QM without a mechanical explanation. In the FFGFT, measurement is a *geometric interaction*: The measuring device drives the mode to the nearest stable extremum ($z = +1$ or $z = -1$). This is a deterministic movement in mode space — but because of the unknown initial phase θ and smallest variations of the initial conditions, statistically indistinguishable from Born rule behavior: [B]

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0). \quad (2.2)$$

The two formalisms yield identical statistical predictions (Doc. 230). Interpretively they differ: What QM describes as chance is, in the FFGFT, ignorance of the exact carrier configuration.

2.3 The Absence of the Time Operator

Pauli's objection shows that for a Hamiltonian bounded from below, there cannot be a self-adjoint, canonically conjugate time operator. [E] Time is missing in the Hilbert space — not as an oversight, but structurally.

In the FFGFT, this is not a deficiency: $\tilde{T}$ is not an operator, but a mode-bound property of the carrier. For mode i , $\tilde{T}_i = 1/m_i$; different modes carry different time scales. Superposition of modes of different mass thus results in *incompatible time scales* — a structural statement about the limits of cross-mode superposition. [B] (Doc. 334)

The Schrödinger equation remains fully valid as an effective description; the time t in it is the global coordinate that is unrolled from the carrier (Type I pullback, Doc. 333).

For all readers: *That QM has no time operator sounds technical — but it means: The theory can say nothing about the duration of processes, only about states at fixed points in time. The FFGFT makes this explicit: Each particle has its own internal clock ($\tilde{T} = 1/m$). Two particles of different mass literally vibrate in different rhythms.*

Sources: Doc. 230, 307, 322, 327, 333, 334

Chapter 3

Entanglement Without Action at a Distance and Apparent Instantaneity

3.1 What Entanglement Is in the FFGFT

Two qubits are called entangled if their joint state cannot be written as a product $|\psi_1\rangle \otimes |\psi_2\rangle$. The maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (3.1)$$

has a geometric characterization in the FFGFT: The two modes share a common carrier $T^4/\mathbb{Z}_3$. Their (z_i, r_i, θ_i) coordinates are not independent — they are knots of the same topological configuration that cannot be reduced to product form. [B]

Entanglement in the FFGFT is a topological connection, not a nonlocal correlation. The particles never had two separate carriers; they share the same one from the beginning. What QM describes as action at a distance is a property of the common geometry.

3.2 EPR and Locality

The EPR paradox (1935): If QM is complete, entangled particles are nonlocally connected; Einstein considered this unacceptable and postulated hidden variables. [E]

The FFGFT resolves the paradox geometrically. Locality holds on the torus in the sense that no signal is transmitted faster than light. The correlation is a property of the common geometry, not a signal transmission. What is given up is only the idea that each particle would have an *independent* carrier — and that is not observable. [B]

3.3 Apparent Instantaneity: Not a Physical Problem

The collapse of the wave function appears to happen instantaneously: One measures qubit A and B immediately knows which state it is in, no matter how far away. This seems to contradict the theory of relativity.

Before explaining the FFGFT resolution, a step back is worthwhile. In standard mathematics, arithmetic operations are considered instantaneous: If $2 + 2 = 4$, then that holds simultaneously for all places in the expression — no one associates a physical instantaneity with this that would be causally problematic. The equality is a structural statement about the object, not an event in space. The same applies to the state description in QM: The wave function $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is a mathematical description of a state — not a process that propagates anywhere. The instantaneity of the descriptive adjustment after the measurement is mathematical, not physical in nature.

In the FFGFT, the same argument applies at the level of the carrier: The relation $\tilde{T} \cdot m = 1$ is a **structural property** of each mode on $T^4/\mathbb{Z}_3$ — analogous to the fact that an equation holds simultaneously for all its variables. It is a **local constraint** at the same spatial point, not a signal between distant points.

The resolution comes from the separation of two levels: A change at $\mathbf{x}_1$ then propagates via the dynamic field equations: [B]

$$\text{Change at } \mathbf{x}_1 \rightarrow \text{propagation with } v \leq c \rightarrow \text{effect at } \mathbf{x}_2. \quad (3.2)$$

The time delay is $\Delta t = |\mathbf{x}_2 - \mathbf{x}_1|/c$.

Local adjustment and global propagation are two different processes. The local adjustment ($T \cdot m = 1$ at the same point) occurs on the Planck time scale $t_p \sim 10^{-43}$ s — practically instantaneous for every measurable time scale. The propagation to distant points

is limited by the speed of light. This hierarchy explains why many quantum processes appear to be instantaneous.

3.4 The Field Equation and Its Causal Structure

The FFGFT field dynamics follows a wave equation:

$$\frac{\partial^2 \tilde{T}}{\partial t^2} = c^2 \nabla^2 \tilde{T} + Q(\tilde{T}, m, \rho), \quad (3.3)$$

with the source function $Q = -4\pi G \rho \cdot \tilde{T}$. This is a hyperbolic differential equation, characteristic of wave propagation with finite speed.

The causal structure is manifested by the Green's function:

$$G(\mathbf{x}, \mathbf{x}', t - t') = \theta(t - t') \cdot \frac{\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))}{4\pi|\mathbf{x} - \mathbf{x}'|}. \quad (3.4)$$

The Heaviside function $\theta(t - t')$ guarantees causality: No effect before its cause. The delta function encodes propagation with the speed of light. This structure is fully compatible with the special theory of relativity. [B]

A concrete example: If $m(\mathbf{x}_0)$ changes at $t = 0$, then $\tilde{T}(\mathbf{x}_0)$ adjusts immediately (local constraint); at the same time $\nabla^2 \tilde{T} \neq 0$ arises, which generates a field disturbance; this propagates as a wave with $v = c$ and reaches $\mathbf{x}_1$ at time $t = r/c$.

3.5 The Hierarchy of Time Scales

The apparent instantaneity results from the extreme separation of different time scales:

Process	Time Scale	Character
Local T-E adjustment	$t_P \sim 10^{-43} \text{ s}$	immeasurably fast
Wave function evolution	$\sim 10^{-15} \text{ s}$	quantum regime
Causal field propagation	$t = r/c$	relativistic
Macroscopic measurement	$\gg 10^{-9} \text{ s}$	classical

Processes on the Planck scale are so fast that they could not be resolved in time with any conceivable technology. What appears as instantaneous runs on this scale — and is thus causally fully compatible.

3.6 The Complete Duality: Time, Mass, Energy, and Length

The FFGFT describes a comprehensive system of dualities in which all fundamental quantities are connected by $\tilde{T} \cdot m = 1$:

$$\tilde{T} \cdot E = 1 \quad (\text{time-energy duality}) \quad (3.5)$$

$$\tilde{T} = 1/m \quad (\text{time-mass relation}) \quad (3.6)$$

$$E = mc^2 \quad (\text{mass-energy equivalence}) \quad (3.7)$$

$$\ell = \hbar/(mc) \quad (\text{Compton wavelength as length-energy bridge}) \quad (3.8)$$

At the Planck scale, all these dualities converge; there, time, length, mass, and energy are no longer independent:

$$t_P = \ell_P/c = \sqrt{\hbar G/c^5} \approx 5,4 \times 10^{-44} \text{ s}. \quad (3.9)$$

The dualities are not merely mathematical formulations, but have physical consequences: They show that the seemingly different concepts of time, space, mass, and energy are different aspects of the same geometric reality. Depending on the scale, a different aspect dominates: At atomic scales the time-energy duality, at macroscopic scales mass, at cosmological scales the length-time relation.

3.7 Topological Connection: The Current State

The early FFGFT documents (autumn 2025) attempted to explain the Bell correlations by a ξ damping that pushes the CHSH value slightly below the Tsirelson bound. This approach is superseded by Doc. 230 (May 2026). The current position:

Entangled particles share a carrier $T^4/\mathbb{Z}_3$ without action at a distance. The statistical predictions are identical with quantum

mechanics. [B] There is no action at a distance, because there is no separate carrier. The topological connection is not an effect over distance, but a geometric fact of the common carrier. A small ξ deviation $\Delta\text{CHSH} \approx \xi/(2\pi) \approx 10^{-5}$ per measurement is open as a working hypothesis, but no longer the core argument. [S]

For all readers: Two points on the same torus are connected — not by a signal between them, but because they share the same surface. That is the difference: EPR asked about a connection between two separate objects. The FFGFT says: The premise is wrong. The objects were never separate.

Sources: Doc. 131 (Apparent Instantaneity, 2026), Doc. 230 (Entanglement as topological connection, May 2026), Doc. 334, Doc. 364

Part II

Bell's Inequality

Chapter 4

Bell's Theorem

4.1 The Fundamental Question

Bell tests check whether the correlations of entangled particles can be explained by a local-realistic theory — a theory in which each particle has properties defined from the beginning that are measured independently of the other particle.

John Bell showed in 1964 that no such theory can reproduce the quantum predictions. [E] The argument is mathematically sharp and cannot be circumvented by technical refinements.

4.2 The CHSH Inequality

In the form formulated by Clauser, Horne, Shimony, and Holt (CHSH, 1969), the Bell inequality reads: [E]

$$|E(a, b) - E(a, c)| + |E(a', b) + E(a', c)| \leq 2, \quad (4.1)$$

where $E(a, b)$ denotes the correlation of the measurement results at the detector angles a and b .

Quantum mechanics violates this bound up to the Tsirelson bound $2\sqrt{2} \approx 2,828$. [E] All loophole-free experiments since 2015 confirm the QM prediction and refute local-realistic theories. [E]

4.3 The Three Assumptions

Bell's proof assumes three assumptions:

1. **Locality:** The result at detector A does not depend on the setting at detector B.
2. **Realism:** The measurement results are predetermined (by hidden variables).
3. **Freedom of choice:** The detector settings are chosen independently of the hidden variables.

At least one of these assumptions is false — that is Bell's achievement. QM gives up locality and realism together.

4.4 Historical FFGFT Approaches and Their Current Status

The early FFGFT documents (Doc. 023, 023a, 074, autumn 2025) attempted various ways to reconcile the violation with local reality:

- **Time field damping** (Doc. 023): The correlations are attenuated by a ξ -dependent factor. Result: CHSH reduction from 2,828 to 2,827 — too small for today's experiments. [S]
- **Superdeterminism** (Doc. 074): The freedom of choice is violated because the energy field jointly determines detector settings and particle properties. This path is superseded by Doc. 230 (see Chapter 5).
- **Topological connection** (Doc. 230, May 2026): No concession to the assumptions necessary — the common carrier makes the correlation a geometric property. This path is authoritative today. [B]

Sources: Doc. 023, 023a, 074, 190 (Prec. 11, 12), 230

Chapter 5

The FFGFT Answer to Bell

5.1 Topological Connection Instead of Action at a Distance

The current FFGFT position (Doc. 230, May 2026) does not challenge any of the three Bell assumptions. Instead, it changes the ontological framework: Entangled particles have no separate carrier, but share the same torus $T^4/\mathbb{Z}_3$. The correlation is a topological property of this common carrier.

This means concretely:

- **Locality** holds in the sense of the carrier: No signal transmits information faster than light.
- **Realism** holds in the sense that the mode configuration exists before the measurement.
- **Freedom of choice** is not restricted.

What is given up is the idea that each particle would have an *independent* carrier. The topological connection is not action at a distance — it is a geometric fact. [B]

5.2 Statistical Predictions

The FFGFT reproduces the Born rule and thus all statistical predictions of QM completely: [B]

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0). \quad (5.1)$$

In particular, the CHSH bound $2\sqrt{2}$ is reached — not undershot, as the early damping approaches had predicted. The Bell test dispute over local realism resolves geometrically: There is no nonlocality, because there is no separate carrier.

5.3 The ξ Deviation

A small deviation from the Tsirelson bound is derivable from the θ_4 projection geometry of the measurement act. The candidate for the elementary deviation per measurement is: [S]

$$\Delta\text{CHSH}_{\text{elem}} = \frac{\xi}{2\pi} \approx 2,1 \times 10^{-5}. \quad (5.2)$$

This is the phase cost of a θ_4 cycle in geometric units (Doc. 190, Precision 12). Cumulatively over N measurement runs, this deviation grows as:

$$\Delta\text{CHSH}(N) \approx \xi \frac{\ln N}{D_f} \cdot 2\sqrt{2}. \quad (5.3)$$

For $N = 73$ this yields $\approx 5 \times 10^{-4}$ — a different observable than the elementary one, both below NISQ noise ($\sim 10^{-2}$).

Experimental reach: Hardware measurements (IBM Heron r2, May 2026, Doc. 147) show Bell violation $S = 2,74$ (96,9 % of the Tsirelson bound). The ξ effect itself requires sub-ppm-stable error-corrected qubits — not resolvable with today's NISQ hardware. [K]

For all readers: *The FFGFT says: The strong quantum correlations are real and correct — QM is right. What it sees differently: The correlation does not come from a connection between the particles, but because the particles were never really separate. They sit on the same geometric carrier like two points on the same drum.*

Sources: Doc. 190 (Prec. 11, 12), 230, 353, 364, 365

Chapter 6

No-Go Theorems and What They Show

No-go theorems limit what alternative interpretations of QM can achieve. The FFGFT must be measured against them.

6.1 Bell's Theorem (Revisited)

Bell's theorem (Chapter 4) excludes local-realistic theories with hidden variables. The FFGFT circumvents this not through a loophole, but through a different ontology: The "hidden variable" is the complete carrier configuration $(z, r, \theta)_i$ of all modes — but the carrier is shared, not locally separated.

6.2 Kochen-Specker Theorem

The Kochen-Specker theorem (1967) shows: No definite values can be assigned to a quantum system for all observables simultaneously and context-independently. [E] Measurement results are context-dependent.

In the FFGFT: The pole transition during measurement depends on the complete carrier-apparatus interaction, not only on the mode itself. Context-dependence is thus a property of the measurement dynamics on the carrier, not a proof of incompleteness of the

description. [S] The Born rule is reproduced; whether the context-dependence is fully derivable from the geometry remains open.

6.3 PBR Theorem

The Pusey-Barrett-Rudolph theorem (2012) shows: If the quantum state were only epistemic (expression of knowledge, not of reality), certain measurements would allow results that QM excludes. [E]

In the FFGFT, the state (z, r, θ) is a real carrier configuration, not a state of knowledge — the PBR theorem is thus compatible: The FFGFT has an ontic wave function. [B]

6.4 Hardy's Theorem and GHZ State

Hardy (1992) and Greenberger-Horne-Zeilinger (1989) construct scenarios in which every local-realistic assignment leads to a contradiction — even without statistical inequalities. [E]

Both arguments presuppose separate carriers. In the FFGFT this presupposition is dropped: The modes share a carrier, and the GHZ correlations are topologically encoded. [S]

6.5 What the FFGFT Does Not Claim

The no-go theorems show what is not possible. The FFGFT claims:

- No new predictions in standard Bell tests (all statistical results as in QM). [B]
- A small ξ deviation in high-precision tests ($\Delta\text{CHSH}_{\text{elem}} \approx 2 \times 10^{-5}$), not measurable today. [S]
- No violation of freedom of choice (no superdeterminism — this approach from Doc. 074 is superseded).
- No information faster than light.

Sources: Doc. 055, 074, 230

Chapter 7

CHSH Deviation and Experimental Reach

7.1 Two Different Observables

The correction register (Doc. 190, Precision 11 and 12, May 2026) distinguishes precisely between two different quantities that earlier documents conflated:

Elementary deviation. Per individual measurement, phase costs of the θ_4 projection cycle arise. The natural candidate is:

$$\Delta\text{CHSH}_{\text{elem}} = \frac{\xi}{2\pi} \approx 2,1 \times 10^{-5}. \quad (7.1)$$

This identification is a working hypothesis; a formal derivation from the T^4 projection geometry is still missing. [S]

Cumulative deviation. Over N measurement runs, the deviation grows according to the formula from Doc. 022:

$$\Delta\text{CHSH}(N) = 2\sqrt{2} \left[1 - \exp\left(-\xi \frac{\ln N}{D_f}\right) \right] \approx \xi \frac{\ln N}{D_f} \cdot 2\sqrt{2}. \quad (7.2)$$

For $N = 73$ this yields $\approx 5 \times 10^{-4}$. This quantity is experimentally accessible once error correction thresholds are surpassed.

7.2 Experimental Situation 2026

IBM Heron r2 (May 2026, Doc. 147) provides:

- Bell violation confirmed: $S = 2,74$ (96,9 % of the Tsirelson bound). [K]
- FFGFT circuits operational on current hardware. [K]
- Device-to-device variation: 1,4 % — approximately $1400\times$ larger than the sought ξ effect. [K]

The ξ effect requires sub-ppm-stable error-corrected qubits. Not resolvable with NISQ hardware. [K]

7.3 Open Derivation

The identification $\Delta\text{CHSH}_{\text{elem}} = \xi/(2\pi)$ is plausible, but not derived from the T^4 projection geometry. The ratio between cumulative and elementary deviation for N runs is:

$$\frac{\Delta\text{CHSH}(N)}{\Delta\text{CHSH}_{\text{elem}}} \approx N \cdot \frac{2\pi}{D_f}, \quad (7.3)$$

for $N = 73$ a factor ≈ 153 , consistent with the observed factor $\sim 50\text{--}100$. [S]

The formal derivation of $\xi/(2\pi)$ from the carrier geometry remains an open bridge.

Sources: Doc. 022, 147, 190 (Prec. 11, 12), 230

Part III

Quantum Computing

Chapter 8

More Than Two States: The Qubit Geometrically

8.1 What Lies Between $|0\rangle$ and $|1\rangle$?

Spin-up and spin-down are two points. Two points are uniquely labeled — one bit of information. But a qubit is not a bit. The actual power of the qubit lies not in the fact that it can be $|0\rangle$ or $|1\rangle$; a classical coin can do that too. It lies in the fact that it can superpose both in a continuous ratio:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (8.1)$$

Two parameters, the mixing angle $\theta \in [0, \pi]$ and the relative phase $\phi \in [0, 2\pi)$, parameterize a spherical surface: the Bloch sphere. On it there are infinitely many points — not only the poles $|0\rangle$ and $|1\rangle$, but the entire equator, all latitude circles, every point in between. Each is a valid, physically distinguishable state.

A qubit encodes not 2 states, but infinitely many

The two classical poles are special cases of a continuous family. What distinguishes a qubit from a bit is not the number of discrete states, but the continuous geometry of the state space. Readout couples to the configuration and drives it into the nearest stable extremal state. The result is, in the FFGFT view, deterministic — it depends on the exact initial state, which we do not fully know before the measurement.

The states themselves are discrete; the continuity lies in the phase space *before* the measurement.

8.2 The Cylindrical Phase Space

The Bloch sphere is the standard representation. The FFGFT has a physically more direct alternative: the cylindrical phase space with three coordinates (z, r, θ) . Here $z \in [-1, +1]$ is the height along the cylinder axis — the proportion of the torsion configuration in the direction of $|0\rangle$ versus $|1\rangle$; $r \in [0, 1]$ the radius in the cross-section — the strength of the intermediate torsion, i.e., how far the state is from both poles; $\theta \in [0, 2\pi)$ the azimuthal angle — the relative phase between the basis components. The relationship to the Bloch sphere is direct: $z = \cos \theta_{\text{Bloch}}$, $r = |\sin \theta_{\text{Bloch}}|$; the formal bridge is in Chapter 13.

The cylinder is not a geometric aesthetic device: It makes gate operations into geometric transformations in real space.

Gate	Matrix	Cylinder Transformation
Hadamard H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Swap $z \leftrightarrow r$, rotate phase by $\pi/2$: moves pole to equator
Phase Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Add π to phase θ : rotation about the cylinder axis
Bit-Flip X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	(z, r) rotation by $\alpha = \pi K_{\text{frak}}$

In the cylindrical formalism, every gate operation is a geometric transformation of the torsion vector. The Hadamard gate does not rotate a matrix — it tilts the configuration from the longitudinal axis onto the transverse plane. The Z gate rotates the winding about its own axis. The factor K_{frak} in the X gate is not a subsequently inserted correction factor; it follows from the fractal space-time geometry and means: Every qubit rotation is slightly smaller than π , because space-time itself deviates minimally from flat geometry. This is a

testable prediction: All π gates should systematically deviate from the ideal value by $\Delta\alpha \approx 0,013\%$. [S]

8.3 Quantum Registers: Exponential Growth Through Combination

A single qubit has infinitely many intermediate states, but on readout always yields one of the two poles. The actual power comes through combination.

Product states: additive classical. Two independent qubits together have four basis states. The general state without entanglement is $(\alpha_A|0\rangle + \beta_A|1\rangle) \otimes (\alpha_B|0\rangle + \beta_B|1\rangle)$ — two separate Bloch spheres, no gain over two independent qubits. The state space is additive: $2 + 2 = 4$ parameters.

Entangled states: multiplicative growth. As soon as the qubits are entangled, the joint state can no longer be written as a product. The general two-qubit state

$$|\Psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \quad (8.2)$$

requires four complex amplitudes, i.e., $2 \cdot 4 - 2 = 6$ real parameters. Entangled qubits carry joint information that is in none of the individual qubits. For n qubits the Hilbert space grows exponentially, $d(n) = 2^n$, with $2 \cdot 2^n - 2$ real parameters:

n Qubits	Basis States	Real Parameters	Classical
1	2	2	1 Bit
2	4	6	2 Bits
3	8	14	3 Bits
10	1024	2046	10 Bits
50	10^{15}	2×10^{15}	50 Bits
300	10^{90}	2×10^{90}	$\approx$ atoms in the universe

That is the actual quantum advantage: Not a single qubit is powerful, but the collective growth of an entangled register.

Bell states. The four maximally entangled two-qubit states form an orthonormal basis of the entangled subspace:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad (8.3)$$

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle). \tag{8.4}$$

If one measures qubit A, the result of qubit B is determined, no matter how far away. In the FFGFT picture, a Bell state is not action at a distance between two qubits, but a topologically connected torsion configuration across two resonators: Both share a common torsion knot, whose resolution upon any local measurement determines both ends simultaneously. This is not a signal with superluminal speed, but a common geometric state that was established during preparation (Chapter 3). The earlier damping formula for the Bell correlation, which lowered the CHSH value by $\sim 10^{-4}$, is replaced by the two-observable reading from Chapter 7.

Hardware validation. The FFGFT predictions for Bell states were tested on IBM Brisbane and IBM Sherbrooke (127 qubits each, Heavy-Hex lattice): Bell state fidelity 97,17 %; FFGFT algorithm and standard QM agree within measurement accuracy; repeat variance 10^{-4} to 10^{-3} ; deviations in $P(00)$ and $P(11)$ of 1–5 % are hardware noise. [K] The ξ deviation from the ideal Bell correlation lies below the noise threshold and could not be isolated. The complete Python code is available in the repository.

8.4 Three Ways to More States

From resonator physics, three fundamentally different strategies emerge to encode more than two states:

Strategy	More States Through	Growth	FFGFT Meaning
Superposition (1 qubit)	continuous superposition	∞ , but 1 bit on measurement	intermediate torsion
Register (n qubits)	entanglement	2^n	topologically connected knots
Qudit ($d > 2$)	higher levels, orbitals	d per object	d stable configurations in one resonator

No strategy dominates everywhere. Register entanglement scales most strongly, but requires n physical objects and error-corrected entanglement operations. Qudit encoding gains information per object ($\log_2 d > 1$); two ququarts ($d = 4$) carry $4^2 = 16$ states, four qubits $2^4 = 16$ — mathematically equivalent, physically differently robust. Superposition is always there: It is the basis of interference, the actual driving force of quantum algorithms.

8.5 Quantum Interference: Why Superposition Is Useful

Superposition alone would be a mere superposition; readout would always yield one of the two poles. What makes superposition a resource is interference: Amplitudes can reinforce constructively or cancel destructively, depending on phase. [E] This is the principle behind all useful quantum algorithms: One prepares the register in a uniform intermediate torsion over all inputs (Hadamard); applies a function that increases the phases of the correct answers and decreases those of the wrong ones; lets interference drive the wrong answers into unstable configurations and the correct one into a stable one; and readout drives the system into the nearest stable pole, which after the interference corresponds with overwhelming frequency to the sought answer. Without interference: Monte Carlo. With interference: Grover ($\sqrt{N}$ instead of N steps), Shor (factorization in polynomial time).

In the FFGFT, the phase is not an abstract complex number, but the rotational position of the intermediate torsion configuration. Constructive interference: Two torsion phases lie equal and reinforce each other into a geometrically more stable configuration. Destructive interference: Two opposite phases cancel each other out; the resulting configuration has no stable geometric anchor and is not driven on readout. A quantum algorithm is, in this language, a sequence of torsion rotations that align the phase configuration of the register so that the sought answer ends in a stable configuration.

For all readers: *Water waves that meet reinforce each other when crest meets crest, and cancel each other out when crest meets trough. A quantum computer controls probabilities like waves: It lets the correct answers grow and the wrong ones disappear. This only works because quantum states have a phase that classical bits lack.*

8.6 QND Measurement: Readout Without Destroying

The readout interaction drives the intermediate configuration into a pole — for invasive measurement methods. There is a class of methods that do not perform this step: the quantum non-demolition measurement (QND). It reads out an observable $\hat{A}$ without changing the eigenstate; the formal condition is $[\hat{A}, \hat{H}_{\text{measurement}}] = 0$. [E] Practically: The same state can be measured repeatedly and yields the same result each time.

The most important example is Faraday or Kerr rotation: A linearly polarized laser beam passes through the quantum dot without absorption and rotates its polarization plane by $\theta_F \propto S_z$. No photon is absorbed, no energy transfer into the spin mode; the spin state is identical before and after the measurement.

QND as empirical evidence for determinism. The argument is direct: The Faraday rotation measures S_z without changing the state; repeated measurements yield the same result every time; therefore the state had this value before every measurement, definitely, not as an abstract superposition that only becomes reality through measurement. In the FFGFT, the Faraday rotation is state-preserving because a stable torsion configuration cannot be driven out of its state by a dispersive, non-resonant coupling. The photon sees the torsion — it rotates its polarization accordingly — but transfers no energy, because it is not resonant. [S]

Method	Interaction	State After
Resonant fluorescence	Photon absorbed, excites transition	changed
Spin-to-charge conversion	Electron tunnels out of the resonator	destroyed
Faraday/Kerr	Photon passes dispersively	unchanged: QND

The standard reading, that the state only exists through measurement, is difficult to reconcile with QND measurements. In the

FFGFT, the torsion configuration is a geometric reality: stable, definite, pre-existent. Measurement reads it off; it does not create it.

Limits of QND. QND is possible for observables that commute with the measurement operator, typically S_z . An intermediate configuration — a state that is not an S_z eigenstate — is driven into an eigenstate by an S_z QND measurement, deterministically depending on the initial state; the intermediate torsion is then gone.

8.7 The Potential Landscape

The complete picture of qubit states is a potential landscape with two stable minima and a saddle in between. The minimum $|\uparrow\rangle$ is a stable configuration, deep enough that weak couplings cannot lift the state out of it — QND possible. The minimum $|\downarrow\rangle$ is mirror-symmetrically equally deep. The slope is the intermediate torsion: The phase determines how far the state is from which minimum; not a true minimum, metastable on the coherence time T_2 . Every point in between is a geometrically defined state, continuously parameterized, discrete in the endpoints.

The readout strength decides: A state in the stable minimum remains under weak coupling (QND) and can be lifted out under strong coupling (invasive). A state on the slope rolls into the nearest minimum even under weak coupling — which one, the exact phase determines, epistemically unknown, deterministic; under strong coupling the same, only faster.

The crucial insight: There is no principled boundary between “QND possible” and “QND impossible”, only more stable and less stable states in a continuous landscape. The jump into the final position is not random: A state near $|\uparrow\rangle$ lands in $|\uparrow\rangle$, one near $|\downarrow\rangle$ in $|\downarrow\rangle$, one on the saddle depending on the fine phase. The apparent randomness of the standard QM is the epistemic ignorance of this phase. The picture holds for all d levels of a qudit: d minima with $d - 1$ saddle points.

8.8 Two Levels of the ξ Hierarchy: Spin Qubit and Photonic Resonator

The previous sections developed the qubit on the spin qubit — an electron in a nanometer cage. There is a qualitatively different realization class: the photonic resonator. Both enforce discrete states through geometry, but sit on different levels of the ξ hierarchy, with consequences for temperature, stability, and the nature of information.

Spin qubit: minimal resonator at level $N \approx 8$. A single electron, one energy level, two orientations. The separation of the spin states in a typical field of 1 T is $\Delta E_Z = g^* \mu_B B \approx 0.1\text{--}1$ meV; the thermal energy at room temperature is $k_B T = 26$ meV. Thus $\Delta E_Z \ll k_B T$: Thermal noise constantly throws the system out of the minimum. Deep temperatures below 1 K are not optional, but mandatory — not to preserve superposition, but to remain in one of the minima at all.

Photonic resonator: collective geometry at level $N \approx 9\text{--}10$. A waveguide of lithium niobate, a Mach-Zehnder interferometer, a ring resonator — resonators of 10^{18} to 10^{23} atoms in lattice order. The resonator is not a single particle, but the macroscopic geometry of the entire crystal; the photon belongs to the collective mode. The discreteness comes from the boundary conditions of the macroscopic geometry. The energy of an optical photon is 1–2 eV, thus $E/k_B T \approx 40$: Thermal noise cannot create optical photons. Photonics works at room temperature not because it is built more robustly, but because it operates on a higher ξ level: The geometric robustness is given by the single crystal with 10^{23} lattice sites, the thermal by the eV scale — four orders of magnitude apart.

Not limited to two states. In a Mach-Zehnder interferometer, the photon runs through two paths with continuously adjustable phase; the interference result $\cos^2(\phi/2)$ is a continuous transfer function. A photonic resonator can operate both modes: digital with $\phi \in \{0, \pi\}$ and transmission 1 or 0, or analog with continuous phase — infinitely many intermediate states, all stable at room temperature. In the spin qubit, the analogous information lies in a thermally unstable intermediate state, T_2 collapses in nanoseconds at room temperature; in the photonic resonator, it lies in the phase of the photon, and that is thermally stable.

	Spin Qubit	Photonic Resonator
ξ level	$N \approx 8$ (nm)	$N \approx 9-10$ ($\mu\text{m}-\text{mm}$)
Resonator	1 electron in cage	$10^{18}-10^{23}$ atoms collective
Energy scale	0,1–1 meV	1–2 eV
$\Delta E/k_B T(300\text{ K})$	0,004–0,04	≈ 40
Cooling needed?	yes, $T < 1\text{ K}$	no
Analog states	fragile ($T_2 \sim \text{ns}$)	stable
Discreteness from	confinement (1 particle)	mode geometry
Typical application	digital qubit	analog signal processing, photonic computing

Spin qubit and photonic resonator are, in the FFGFT, not fundamentally different systems, but resonators at different levels of the same scaling ladder. Each level carries the energy by the factor $1/\xi \approx 7500$ higher than the previous one; from meV (level 8) to eV (level 9–10) the factor is 10^3-10^4 , consistent with the scaling. [5] This also explains why biological systems use photonic mechanisms (photosynthesis, bird navigation) and no spin qubits for quantum effects: Nature works on the thermally robust level.

8.9 The Analog Resolution Limit

The photonic resonator is thermally robust, but only as long as one does not resolve too finely. Anyone who wants to distinguish analog intermediate values divides the energy space into N levels with $\Delta E_{\text{level}} = E_{\text{photon}}/N$, and the condition $\Delta E_{\text{level}} \gg k_B T$ yields $N \ll E_{\text{photon}}/k_B T$. At room temperature and optical photons: $N_{\text{max}} \approx 40$, corresponding to $\log_2 40 \approx 5,3$ bit per photon. The photonic system does not escape the thermal problem — it shifts it to a finer scale. More resolution requires cooling or an even higher energy scale.

This is the oldest engineering wisdom of communications engineering, geometrically grounded: High-frequency technology has

been climbing the same hierarchy for a hundred years. At each frequency level, the signal-to-noise ratio improves because $E/k_B T$ grows with each level.

Technology	Frequency	$E/k_B T$	N -Level	Regime
AM radio	1 MHz	$1,5 \times 10^{-7}$	$\approx 5,5$	thermally dominated
FM	100 MHz	$1,5 \times 10^{-5}$	$\approx 6,4$	noise-limited
Microwave	10 GHz	$1,5 \times 10^{-3}$	$\approx 7,3$	manageable
5G/mmWave	30 GHz	5×10^{-3}	$\approx 7,5$	transition
THz	1 THz	0,15	$\approx 7,8$	$E \sim k_B T$: difficult
Near-infrared	200 THz	31	$\approx 8,8$	thermally robust
Fiber optic	300 THz	46	$\approx 9,0$	$N_{\max} \approx 46$
UV	1000 THz	154	$\approx 9,4$	$N_{\max} \approx 150$

From $N = 6$ (AM) to $N = 9$ (optics), three levels bridge the factor $(1/\xi)^3 \approx 4 \times 10^{11}$; in fact, the fiber optic frequency is 10^8 -fold above AM radio. The fiber optic is, in this picture, not a new technology, but the continuation of the same trend to the next level. And the resolution limit applies at every level: Anyone who wants to resolve more than 40 analog levels at optical wavelengths must cool — as the radio engineer in 1930 shielded his receiver. The universal limit reads

$$N_{\max}(T, N_\xi) \approx \frac{E_0 (1/\xi)^{N_\xi - N_0}}{k_B T}. \quad (8.5)$$

Above optics. Optics is not the end of the ladder, but the end of the usable zone for guided photonic transmission. In the near UV (3–10 eV, $N \approx 9,3$) photoionization begins: The resonator is damaged by its own signal. At soft X-rays (100 eV–1 keV, $N \approx 10$) the wavelength approaches the lattice spacing, Bragg scattering at individual lattice planes, the waveguiding collapses. At hard X-rays, the photon flies through the lattice gaps, matter becomes partially transparent, there is no refractive index and no resonator. At gamma energies, the photon interacts only with atomic nuclei — another ξ level. Above 1 MeV ($N \approx 11,3$) pair production sets in: the natural upper limit. In the FFGFT there are no cannonballs and no waves as separate concepts, only torsion configurations at various scales;

waveguiding is a scale resonance of two torsion patterns, the crystal at level 7–8 and the photon at level 9, and it breaks down when the scales become incompatible.

What This Chapter Holds

The qubit is a mode in the cylindrical phase space; gates are torsion rotations; registers grow exponentially through topological connection; interference is phase superposition of torsions; QND measurement evidences the pre-existence of stable poles; the potential landscape makes the measurement outcome deterministic with epistemically unknown phase. Spin qubit and photonics are resonators at different ξ levels, and the thermal resolution limit applies equally at every level.

Sources: Doc. 175 (Qubit state spaces, March 2026), Doc. 034 (cylindrical formalism), Doc. 173 (resonator conditions), Doc. 174 (torsion, Faraday readout), Doc. 230

Chapter 9

Quantum Algorithms: Deutsch, Grover, Shor and the Limits of the Period

9.1 The Four Standard Algorithms

Four algorithms form the textbook lesson of quantum computing. They show what quantum interference can achieve and what it cannot achieve.

Deutsch: One Bit in One Query

The Deutsch algorithm answers the question of whether an unknown function $f : \{0,1\} \rightarrow \{0,1\}$ is constant or balanced, with a single query — classically one needs two.

In standard QM: One prepares $|0\rangle$, applies Hadamard, queries the oracle U_f , applies Hadamard, and measures. Result 0: constant; result 1: balanced. The trick is phase kickback: $U_f|x\rangle = (-1)^{f(x)}|x\rangle$. The Hadamard afterwards makes the global phase difference visible.

In the FFGFT: The same sequence on the cylindrical phase space. The oracle rotates the phase of the torsion vector; Hadamard swaps z and r and rotates by $\pi/2$. The measurement result is identical. There is no algorithmic difference — only the conceptual framework is a different one: The phase is not an abstract complex number, but the rotational position of the torsion configuration. [K]

Grover: Search in $\sqrt{N}$ Steps

Grover searches an unsorted database with N entries in $O(\sqrt{N})$ steps instead of $O(N)$ classically. The mechanism: One prepares the register in a uniform superposition of all N entries, marks the sought entry by phase kickback ($U_{\text{Grover}} : |x\rangle \rightarrow (-1)^{\delta_{x,x_0}} |x\rangle$) and then reflects about the uniform superposition (inversion about the mean). Each Grover step increases the amplitude of the sought entry at the expense of all others. After $\pi/4 \cdot \sqrt{N}$ steps, the probability of the correct entry is near 1.

FFGFT classification: Grover iterations are, in the FFGFT, a sequence of torsion rotations that systematically align the phase of the sought entry. The other entries land in unstable configurations; the readout interaction drives the system deterministically into the aligned pole. [K] The predictions are identical to standard QM.

Shor: Factorization in Polynomial Time

Shor’s algorithm factors N through period finding. Let a with $\text{gcd}(a, N) = 1$; the function $f(x) = a^x \bmod N$ is periodic with period r . The quantum Fourier transform (QFT) makes the period visible in the measurement probabilities; continued fraction approximation extracts r from the measurement value. From r the factors follow via $\text{gcd}(a^{r/2} \pm 1, N)$.

Central statement of the corpus: The algorithm decomposes into two processes that have nothing to do with each other: [K]

	Preparation	Transformation
Task	Modular exponentiation $a^x \bmod N$	Period extraction
Nature	arithmetic problem	wave operation (interference)
Cost	expensive ($O(n^3)$ gates)	cheap ($O(n^2)$ gates)

The preparation accounts for $\approx 95\%$ of the gate costs; the QFT only $\approx 5\%$. [K] Shor specified only the classical process $(a, 1) \rightarrow (a, a^x \bmod N)$ — the implementation of modular exponentiation in superposition is the actual engineering problem and does not depend on quantum mechanical properties.

9.2 ϕ -QFT and Equivalence Theorem

Doc. 147 develops an alternative Fourier transform that uses phases $2\pi/\varphi^k$ instead of $2\pi/2^k$, where φ is the golden ratio:

$$\phi\text{-QFT} : |x\rangle \mapsto \frac{1}{\sqrt{Q_\varphi}} \sum_{y=0}^{Q_\varphi-1} e^{2\pi i xy/Q_\varphi} |y\rangle, \quad (9.1)$$

with $Q_\varphi = \varphi^n$ instead of $Q = 2^n$.

Main theorem: For Shor's algorithm for factorizing $N < 2^{20}$ with error probability $\delta < 10^{-6}$: [K]

$$\begin{aligned} P_{\text{success}}(\text{Standard-QFT}) &\leq P_{\text{success}}(\phi\text{-QFT}) \\ &\leq P_{\text{success}}(\text{Standard-QFT}) + \xi. \end{aligned} \quad (9.2)$$

The ϕ -QFT is equivalent to the standard QFT, with a maximum deviation of $\xi \approx 1,3 \times 10^{-4}$ — negligible. The equivalence theorem shows: ξ has no structural function in $\mathbb{Z}/N\mathbb{Z}$. The period r is determined by $\text{lcm}(p-1, q-1)$, an individual arithmetic property of N , not a universal geometric parameter.

9.3 The Structural Limits of the Period

Three limits set precise bounds on what quantum algorithms can achieve.

1. Weyl law and counting density. [K] The eigenvalue density of every compact state space grows according to a power law $N(\lambda) \sim \lambda^{d/2}$. The prime counting function $\pi(T)$ and the Riemann zeros grow like $T \ln T$. That is not a power. A finite state space cannot fully carry such a spectrum. The Weyl obstruction: An n -qubit register is a compact state space; arithmetic counting densities grow faster than any compact geometry allows.

2. Preparation as bottleneck. [K] The QFT makes periodicities visible that are already present in the prepared states. The limiting step is state preparation — the evaluation of $a^x \bmod N$ — not the transformation itself. No geometric parameter ξ can reduce the arithmetic complexity of the preparation: It is a function of the prime structure of N , not of the geometry of the carrier.

3. No quantum advantage for RSA (as of 2026). [X] An advantage must become visible on the Shor factorization curve: The threshold of error correction must be demonstrated and maintained; the preparation must be executed in superposition; the result must be replicated with RSA-relevant key lengths (2048 bit). None of this is realized as of 2026. Google Willow does surpass the error correction threshold for logical qubits — but in a regime far below the scaling necessary for Shor.

9.4 Riemann Zeta and FFGFT Carrier

Doc. 316 and Doc. 343 clarify the relationship between the Riemann zeta and the FFGFT carrier.

The Riemann zeta as a factor of the torus spectral zeta. [K] The closed form of the $\mathbb{Z}^4$ spectral zeta is:

$$Z_{\mathbb{Z}^4}(s) = 8(1 - 4^{1-s})\zeta(s)\zeta(s-1). \quad (9.3)$$

This is an identity. The zeros of the torus spectral zeta have three sources: the Riemann zeros (from $\zeta(s)$), their shifted ones (from $\zeta(s-1)$) and the pole-zero $s=1$. The Riemann hypothesis is thus a statement about a geometric object of the framework — but one that the geometry cannot answer: The torus can carry the zeros as a factor, but cannot generate them.

$\mathbb{Z}_3$ orbifold projection. [B] The $\mathbb{Z}_3$ projection of $T^4/\mathbb{Z}_3$ transforms the full Riemann zeta into a Dirichlet L -function:

$$\zeta_{T^4/\mathbb{Z}_3}(s) = L(s, \chi_0) = (1 - 3^{-s})\zeta(s). \quad (9.4)$$

The Euler product of this L -function is hierarchically ordered according to the entry depth $k = \text{ord}_p(3)$; the physically visible primes $\{2, 3, 5, 7, 11, 13\}$ are exactly those with $k \leq 6$.

Galois structure and neutrinos. [B] The four Frobenius conjugacy classes of primitive elements in $\text{GF}(27)^*$ are paired by the FFGFT sector pairing $k \leftrightarrow -k$ into two classes: $\{f_3, f_4\}$ (Dirac, charged leptons) and $\{f_1, f_2\}$ (Majorana, ν_1, ν_2 active; ν_3 in Dirac orbit O_4). Prediction: ν_3 has no Majorana phase.

Weyl limit. [K] No compact geometry can carry the zeros of the Riemann zeta as a spectrum. The zero density $N(T) \sim T \ln T$ demands a dilatation structure — the unrolled domain. The geometry shows

the address, but does not provide the answer. Three obvious geometric approaches are excluded: No compact geometry provides the $T \ln T$ growth; ξ does not shift the critical line; the ξ ladder does not discretize the zeros.

9.5 Deterministic Implementation on a PC

The state bridge $(z, r, \theta) \leftrightarrow (\alpha, \beta)$ is bijective (Chapter 13). It follows directly that all logic building blocks — gates, algorithms, state operations — are implementable as exact matrix operations on a normal PC. No quantum substrate, no repetitions, no shot averaging. Every result is fully determined in a single pass.

This was implemented and compared with IBM QC results (pruef_logikbausteine.py):

Gates H , X , Z , CNOT as exact unitary matrices; X_{frak} with rotation angle πK_{frak} , deviation 1.333 %, infidelity 4.4×10^{-4} . One matrix product — no sampling.

Deutsch Constant and balanced oracle: one single run each, exact result.

Grover $n = 3$ to $n = 10$ qubits deterministically; $P(\text{target}) > 0.94$ in one pass without iteration. Scales on a normal PC up to $n \approx 25$ (memory $2^{25} \times 16$ bytes ≈ 512 MB).

Bell state $P(00) = P(11) = 0.5$ exactly in one run. IBM Kingston: fidelity 0.9876 over 50×2048 shots. The QC required 100,000 measurements; the PC: one single pass.

Period finding $N = 15$ to $N = 323$ deterministically, all factors correct. No sampling, no QC. [K]

The deterministic results agree with the IBM QC results within measurement accuracy — not approximately, but exactly. The difference lies solely in the method: the QC measures probabilistically and requires many repetitions; the PC computes the amplitude directly.

9.6 Classical PC versus Quantum Computer

Deterministic single run. [K] The FFGFT state map $(z, r, \theta) \leftrightarrow (\alpha, \beta)$ is bijective (Chapter 13). It follows directly that all algorithmic steps — period finding, factor extraction, gate operations — are fully determined on a PC in a single pass. No repetitions, no shot averaging, no statistics. The IBM Kingston comparison required 50 runs of 2048 shots each on the QC side, because the QC measures probabilistically; the FFGFT simulation on the PC: one single run, identical result.

Preparation: PC faster than QC. [K] Modular exponentiation $a^x \bmod N$ on the PC is direct classical arithmetic with no overhead. The quantum circuit for the same task is demonstrably *more expensive* than the classical implementation, due to reversibility, uncomputation, and Toffoli gates (Doc. 176). Shor never specified the preparation in superposition; it is and remains an arithmetic problem, equally costly in every technology.

Classical methods outperform period search. [K] The FFGFT period search runs sequentially at $O(N)$; the measured growth exponent of the mean order is 0.95. Pollard ρ requires 3–7 steps for the same task, trial division 58–180. The classical resonance search also fails doubly: $O(N)$ steps *and* a frequency resolution of $1/N^2$ that falls below any numerical representation at RSA-relevant sizes (s1_periodensuche.py, Doc. 176).

Entanglement does not carry the computational advantage. [K] Gottesman–Knill (1998): every Clifford circuit (H , S , CNOT) is classically simulable in polynomial time — regardless of the entanglement generated. A GHZ state with $n = 1000$ qubits is maximally entangled and requires only $2n^2$ classical bits of storage. What carries the quantum advantage is unknown; contextuality and non-stabilizerness are candidates, none established.

The sole exception: QFT in superposition. [K] The only genuine quantum advantage lies exclusively in the QFT reading out the period over all x simultaneously in superposition — $O((\log N)^3)$ instead of $O(N)$. A classical PC cannot replicate this. This is the *only* point at which the QC would be faster.

Currently out of reach. [K] RSA-2048 would require 4.1×10^6 physical qubits and 8.6×10^9 gates — a resource limit, in principle surmountable, but in 2026 out of reach by orders of magnitude.

Google Willow crosses the error-correction threshold for logical qubits but lies far below the scaling required for Shor. The QFT exception exists in theory; it is practically irrelevant today.

9.7 Experimental Validation

IBM Kingston (May 28, 2026), 50 independent runs, 2048 shots per run: Bell state fidelity $0,9876 \pm 0,0028$. The repeatability variance is statistically compatible with ordinary quantum shot noise (ratio observed/expected variance: 1,02). An earlier evaluation of 3 runs had erroneously reported 40× determinism relative to QM; that was a sampling artifact. [K]

The ξ correction to CHSH values is $\Delta \sim 10^{-5}$ per measurement, far below NISQ noise. The frequently cited cumulative values for $N = 73$ qubits (Doc. 022 and Doc. 147) and the elementary deviation $\xi/(2\pi)$ are not directly comparable and must not be equated (Chapter 7).

Sources: Doc. 073 (Algorithms in T0, 2025), Doc. 147 §3 (ϕ -QFT equivalence), §4 (Bell tests), §5 (Shor), §8 (IBM validation, May 2026), Doc. 176 (Preparation vs. transformation), Doc. 316 (Riemann, Weyl), Doc. 343 (Zeta Galois)

Chapter 10

Quantum Dot, Spin, and the Limit of Superposition

10.1 What a Quantum Dot Is: Geometry Enforces Discreteness

A quantum dot is an artificial atom — a cavity so small that its geometry enforces discrete energy levels. Not the number of atoms determines the levels, but the shape of the cavity: how the tones of a flute depend on the length of the tube, not on the number of atoms in the metal. The energy levels of the confined particle:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2m_{\text{eff}} R^2}, \quad n = 1, 2, 3, \dots \quad (10.1)$$

The factor π^2 follows from the geometry: A standing wave that vanishes at the boundary enforces $k \cdot R = \pi$ for the fundamental mode. The discreteness comes from the geometry, not from a postulate.

In the FFGFT, an electron is not a point charge in a box, but a stable torsion configuration of the underlying field. The discrete levels are the allowed geometric configurations of this field twist in a bounded space. Between the levels there is no stable configuration; the field exists only in defined patterns, not in arbitrary intermediate states.

The fundamental parameter $\xi = 4/30000$ generates a universal length hierarchy $L_N = L_0 \cdot (1/\xi)^N$ starting at the fundamental length

$L_0 = \xi \ell_p$. Level $N = 8$ corresponds to $L_8 \approx 22$ nm; the excitonic Bohr radius for CdSe lies at $N \approx 7,85$ — the same scale level. The scale of excitons is, in the FFGFT, not an empirical quantity, but geometrically enforced. [S]

N	L_N	Physical Domain
0	$2,15 \times 10^{-39}$ m	Fundamental FFGFT length
1	$1,62 \times 10^{-35}$ m	Planck length
6	$3,8 \times 10^{-16}$ m	Subnuclear
7	$2,9 \times 10^{-12}$ m	Atomic scale
8	$2,2 \times 10^{-8}$ m	Quantum dot and exciton range
9	$1,6 \times 10^{-4}$ m	Cellular scale
10	≈ 1 m	Macroscopic

10.2 Two Conditions for Quantum Dot Behavior

Two conditions must both be fulfilled:

Condition 1 — Geometric perfection (absolute, temperature-independent): The resonator must be precise enough to enforce standing waves. A defective lattice shows no confinement quantum behavior, no matter how small, no matter at what temperature. Condition 1 is the harder one; it cannot be replaced by cooling.

Condition 2 — Level spacing versus heat (size- and temperature-dependent): The spacing $\Delta E = 3E_1$ between the lowest two levels must exceed the thermal energy: $\Delta E > k_B T$.

From Condition 2 follows the maximum resonator size:

$$R_{\text{bulk}}(T) = \pi \hbar \sqrt{\frac{3}{2m_{\text{eff}}k_B T}}. \quad (10.2)$$

For silicon at 300 K: $R_{\text{bulk}} \approx 15$ nm — exactly where classical transistors stopped scaling in 2012.

Material	m_{eff}/m_e	R_{bulk} at			
		4 K	77 K	300 K	310 K
GaAs	0,067	221 nm	50 nm	25,5 nm	25,1 nm
CdSe	0,130	159 nm	36 nm	18,3 nm	18,0 nm
Silicon	0,190	131 nm	30 nm	15,2 nm	14,9 nm

Cooling to 4 K shifts the limit for silicon from 15 nm to 131 nm — a factor 9. This makes macroscopically acting objects into quantum objects: A 50-nm silicon crystal is classical at room temperature, quantized at 4 K. A macroscopic body does not have no levels, but 10^{23} so densely packed that $k_B T$ covers them all. The discreteness does not disappear; it becomes so fine-grained that it appears as band structure.

10.3 Transistors and the ξ Level Table

When the semiconductor industry had to switch to FinFET in 2012, because classical planar transistors had stopped scaling, the channel lengths had fallen below ξ level $N = 8,00$:

Technology	Channel Length	N -Level	Regime
65 nm (2005)	65 nm	8,12	classical
22 nm FinFET (2012)	22 nm	8,00	quantum influ-
			ences begin
10 nm (2016)	10 nm	7,91	tunneling mea-
			surable
5 nm (2020)	5 nm	7,84	QD regime
3 nm GAA (2023)	3 nm	7,78	true QD channel
2 nm (2025)	2 nm	7,73	≈ 10 atomic lay-
			ers

10.4 Biological Quantum Effects

Living cells use quantum effects at 37 °C. Three classes with different conditions: Confinement quantization requires both conditions; tunneling and spin coherence have their own criteria. Condition 1 (geometric precision) applies to all three without exception.

Effect	Cond.1: Ge- ometry	Cond.2: $\Delta E > k_B T$	Mechanism
Photo-synthesis (FMO)	✓ protein shell	borderline ($J/k_B T \approx 0,7$)	exciton coherence, $J \approx 19$ meV
Bird navigation	✓ cryptochrome	no confinement	spin entanglement, relaxation time > reaction time
Enzyme tunneling	✓ active site	no confinement	quantum tunneling through barrier

The biological operating point at 37 °C is not there because of a thermal energy limit; R_{bulk} changes by less than 10 % between 0 °C and 57 °C. What collapses at 42–45 °C is the geometric precision: Proteins denature, the torsion configuration collapses. 37 °C is the point at which biological resonators (protein cavities ~ 2–4 nm) are maximally chemically active without losing their configuration.

10.5 Influencing Spin: Five Methods

Spin-up and spin-down are the absolute minimum of a quantum resonator: two allowed orientations of a torsion configuration. Five methods to rotate this orientation:

Magnetic field (Zeeman effect). The static field B splits the spin states by $\Delta E_Z = g^* \mu_B B$; a resonant high-frequency pulse at the Larmor frequency rotates the spin in a targeted way. In the FFGFT, the field establishes a preferred direction in the torsion field and energetically breaks the symmetry between up and down. [E]

Electric field (EDSR). Electric fields couple via spin-orbit coupling: The electron sees an effective magnetic field in its rest frame, which rotates the spin. Advantage: Gate electrodes generate fields localized to a few nanometers and switch in less than a nanosecond. For integrated qubit arrays, this is the preferred method. [E]

Circularly polarized light. σ^+ photons carry angular momentum $+\hbar$ and transfer it upon resonant excitation to the exciton and thus to the electron spin. Purely optical spin preparation without an external field; readout via the polarization of the fluorescence light. In the

FFGFT: The photon carries its angular momentum as a topological property of the electromagnetic field torsion; upon absorption, it rotates the electron torsion. [E]

Exchange interaction. When two electrons in neighboring quantum dots overlap spatially, their spins couple with $H_J = J(t)\mathbf{S}_1 \cdot \mathbf{S}_2$. By pulsing the gate field (which controls the overlap and thus $J(t)$), one performs targeted two-qubit operations — the basis for spin qubit gates in semiconductor systems (Loss-DiVincenzo, 1998). [E]

Hyperfine interaction. The electron interacts with the nuclear spins of the $\sim 10^4$ – 10^6 surrounding atoms. The fluctuating nuclear spin bath normally destroys the spin polarization in $T_2^* \sim 10$ – 100 ns — a problem. Used in a controlled way, e.g., by dynamical decoupling (rapid pulse sequences that average out nuclear spin fluctuations), T_2 is extended to microseconds. In phosphorus-in-silicon, the nuclear spin itself is used as a highly stable qubit ($T_1 > 1$ s at low temperatures). [E]

10.6 Reading Out Spin: Five Ways

Spin-to-charge conversion. The potential is chosen such that only spin-down is energetically allowed to tunnel out of the quantum dot; the tunneling electron generates a measurable current pulse in the neighboring single-electron transistor. Accuracies $> 99,5\%$ at readout times below $1\mu\text{s}$; sufficient for error-corrected quantum computers according to the surface code threshold. [E]

Pauli blockade. Two neighboring quantum dots, one electron each. If one tries to load both into the same dot, this works for antiparallel spins (both fit into the ground level), but not for parallel spins (Pauli exclusion). Measurable as a conductance difference. In the FFGFT: Two identical torsion configurations cannot occupy the same resonator, not because of a prohibition, but because the same torsion winding in the cavity does not allow a second stable configuration of the same kind. The Pauli principle is a geometric statement.

Faraday and Kerr rotation. A linearly polarized laser beam passes through the quantum dot without absorption and rotates its polarization plane by $\theta_F \propto S_z$. This is a quantum non-demolition

measurement (QND); the spin state is unchanged. Time resolution in the picosecond range; individual spins detectable. Repeated measurements yield the same result. [E]

Resonant fluorescence. The photon is absorbed and excites the transition — invasive, the state is changed or destroyed afterwards.

Imaging methods. For ensembles: MOKE (magneto-optic Kerr effect), time-resolved Faraday rotation, spin noise spectroscopy.

10.7 Decoherence: The Limit of Superposition

Two time scales describe the decay of quantum states. T_1 (relaxation time) is the time until the system has fallen from the excited state into the ground state. T_2 (coherence time) is the time until the phase of the superposition is destroyed; always $T_2 \leq 2T_1$.

System	T_1	T_2	Main Mechanism
Electron spin GaAs	$\sim 1\text{ ms}$	$\sim 10\text{ ns}$	nuclear spin bath
Electron spin Si/SiGe	$> 1\text{ s}$	$\sim 10\text{ ms}$	^{28}Si (nuclear spin free)
Electron spin P:Si	$> 1\text{ s}$	$> 100\text{ ms}$	isotopic purification
Exciton CdSe QD	$\sim 1\text{ ns}$	$\sim 10\text{ ps}$	phonon scattering
Superconductor transmon	$\sim 100\text{ }\mu\text{s}$	$\sim 100\text{ }\mu\text{s}$	dielectric losses

The comparison between exciton ($T_2 \sim 10\text{ ps}$) and spin qubit in Si ($T_2 \sim 10\text{ ms}$) yields nine orders of magnitude. With the Higgs coupling parameter $\xi_{\text{Higgs}} \approx 1,038 \times 10^{-5}$, two hierarchy levels bridge exactly 9,97 orders of magnitude — in agreement with the measured ratio to $< 5\%$. [S] The interpretation: The exciton resonator couples directly to its environment (phonons of the same level), the spin qubit to a distant bath (nuclear spins at level $N \approx 7$), thus two levels weaker.

Silicon enriched with ^{28}Si eliminates the nuclear spin bath almost completely. What remains is the intrinsic geometry of the resonator. The long coherence time in ^{28}Si is evidence that geometric perfection (Condition 1) is the decisive one. [K]

10.8 Superposition: Fragile and Never the Stable Ground State

Here lies the actual conceptual core. The standard theory says: A single electron is in a true superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ until measurement. On every refrigerator there is a permanent magnet. Between these facts lies a tension.

A permanent magnet is the macroscopic sum of spin orientations. The exchange interaction forces neighboring spins into parallel alignment; that is energetically more favorable. This collective alignment forms Weiss domains. In the permanent magnet, the domains are permanently frozen by an external field. What the permanent magnet directly proves: Spin has preferred, stable orientations; this stability is macroscopically robust at room temperature over years; collective superposition never occurs.

The standard theory explains this with decoherence: The interaction with the environment destroys the superposition so quickly that it is practically not observable. This is technically correct, but a subsequent explanation: It begins with superposition as the ground state and then explains why this ground state is always immediately destroyed.

The FFGFT begins the other way around: Stability is the ground state. Superposition is the limited, fragile exception with measurable time scales and a geometric cause.

System	T_1	T_2	Temp.	Remark
Exciton CdSe QD	$\sim 1\text{ ns}$	$\sim 10\text{ ps}$	4 K	Superposition: picoseconds
Electron spin GaAs	$\sim 1\text{ ms}$	$\sim 10\text{ ns}$	mK	Superposition: nanoseconds
Electron spin Si	$> 1\text{ s}$	$\sim 10\text{ ms}$	mK	Superposition: milliseconds
Nuclear spin P:Si	$> 1\text{ s}$	$> 100\text{ ms}$	mK	best known system
Permanent magnet	years	—	300 K	stability without cooling

The table shows: Not a single experimental system shows superposition as a permanently stable state. Even in the best qubit (nuclear spin P:Si at mK), the superposition decays after seconds. The permanent magnet holds its stable spin state over years. Stability is energetically preferred; superposition is metastable.

The single electron. The FFGFT position: A single electron has a definite torsion configuration at every moment — either spin-up or spin-down. What is described as superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ is epistemic ignorance about this definite state, not ontological simultaneity. The proof comes from the QND measurement (Faraday rotation, Chapter 8): A spin state can be measured repeatedly without changing it, and always yields the same result. If the electron were truly in superposition, a QND measurement would yield random results on each repetition. [S]

From single spin to permanent magnet. The transition is, in the FFGFT, not a puzzle, but the same mechanism at different scales: Stability is geometrically grounded, superposition is a metastable intermediate configuration, decoherence is the process that pushes the intermediate torsion into one of the stable extremal configurations. In the permanent magnet, 10^{23} torsion configurations have collectively aligned to the same orientation and stabilize each other. The macroscopic alignment is stable not despite, but because of the geometric coupling.

Sources: Doc. 173 (Quantum dot, March 2026), Doc. 174 (Spin and qubit states), Doc. 178 (Spin stability)

Chapter 11

Hardware: Coherent Ising Machine, Google Willow, p-Bit, Photon Chip

11.1 Coherent Ising Machine: Optimization as Physical Relaxation

The Coherent Ising Machine (CIM) is an optical analog computer that solves combinatorial optimization problems through physical relaxation into the ground state of an Ising Hamiltonian. The fundamental principle: The system *falls* into its solution, it does not calculate it.

The Ising Hamiltonian as Universal Optimization Language

Every NP-hard combinatorial problem can be formulated as minimization of the Ising Hamiltonian:

$$H_{\text{Ising}} = -\frac{1}{2} \sum_{i \neq j} J_{ij} \sigma_i \sigma_j - \sum_i h_i \sigma_i, \quad (11.1)$$

where J_{ij} are the coupling strengths and h_i external fields. The ground state — the spin configuration with minimal H_{Ising} — corresponds to the optimal solution. The mapping is exact, no approximation: Travelling Salesman, MaxCut, portfolio optimization, protein structure prediction, RSA factorization are all formulable as Ising minimization.

Physical Implementation

The CIM implements the Ising model through a network of time-multiplexed degenerate optical parametric oscillators (DOPO): Spin $+1$ is a laser pulse with phase 0° , spin -1 a pulse with phase 180° ; the couplings J_{ij} are realized by electronic feedback. The system evolves deterministically according to the field equations of the parametric oscillator and thereby finds with high probability the ground state. Up to 100,000 optical spins are realized (as of 2024). [E]

A CIM appears to search stochastically; different runs sometimes yield different solutions. But physically the system follows *deterministically* the field equations. The apparent randomness is exclusively ignorance about the exact initial conditions of the quantum vacuum — a direct parallel to the FFGFT measurement.

Six Structural Parallels to the FFGFT

Six parallels can be identified between CIM and FFGFT:

Parallel 1: Determinism behind apparent stochasticity. In the CIM, the solution is physically enforced; in the FFGFT, the measurement result is geometrically determined by the field configuration. Both replace fundamental probabilism with hidden determinism.

Parallel 2: Ground state as geometric minimum. The CIM relaxes into the energy minimum of the Ising Hamiltonian; the FFGFT describes the universe as in the ground state of its own geometry with $\xi = 4/30000$. Optimization is relaxation, not iteration.

Parallel 3: Qubit structure. CIM spin ± 1 is the phase of the laser pulse; FFGFT qubit $|\tilde{T}_\uparrow, m_\downarrow\rangle$ and $|\tilde{T}_\downarrow, m_\uparrow\rangle$ is intrinsically enforced by $\tilde{T} \cdot m = 1$. The FFGFT spin does not need to be externally limited to ± 1 ; it is, through the geometry.

Parallel 4: The coupling matrix. In the classical CIM, the J_{ij} are externally programmed; in an FFGFT-based Ising machine, they would emerge geometrically from the time field: $J_{ij}^{(\text{FFGFT})} = \int \tilde{T}_i \cdot \tilde{T}_j \cdot G(x_i, x_j) d^3x$. This means: no encoding errors possible, since the couplings would be real physical interactions. [S]

Parallel 5: Fractal correction as annealing. The correction factor $K_{\text{frak}}^{3/2}$ plays in the FFGFT the analogous role of the annealing

temperature parameter: It prevents the system from remaining in a local minimum of the geometric energy.

Parallel 6: Toroidal topology. The FFGFT describes the carrier as $T^4/\mathbb{Z}_3$. Toroidal boundary conditions would have an advantage for a physical Ising machine: no boundary problem, all spins have the same number of neighbors, natural implementation of translation-invariant couplings.

11.2 Google Willow: Topological Error Correction and Self-Measurement

Google's Willow quantum processor achieved three breakthroughs in 2024 and 2025 that have special significance from the FFGFT perspective.

Error Correction Below Threshold (December 2024)

Willow (105 superconducting transmon qubits, ~ 10 mK) shows for the first time that the logical error rate decreases exponentially when the code distance d is increased. [E] A $d = 7$ block achieves an error rate by factor 2,14 below the $d = 5$ block. Physical single-qubit error rates are 0,05 % — half as much as on the predecessor Sycamore. The error correction is based on a surface code with topologically protected logical qubits.

FFGFT classification: More physical qubits reduce the logical error exponentially — this is the experimental evidence that topologically stable configurations exist and are reachable in the quantum system, not through external noise suppression, but through geometric order. This is the direction in which the FFGFT points with its mode classes of the D_4 lattice (Chapter 14). [S]

Floquet Topological Order (September 2025)

Will et al. realized on Willow a Floquet topologically ordered phase — a quantum phase that is forbidden in thermal equilibrium, but arises under periodic external driving. [E] On up to 58 qubits arose: chiral edge modes (information propagation in a preferred direction),

anyonic excitations (e-anyons become m-anyons and vice versa), and an interferometrically measurable topological bulk invariant.

FFGFT classification: The non-equilibrium order carries a $\mathbb{Z}_2$ symmetry on the two-dimensional lattice — structurally related to the $\mathbb{Z}_3$ triality of the FFGFT carrier, but not identical (dimension, symmetry group, and carrier geometry different). The Floquet topological order is an experimental analogue of the mode structure on $T^4/\mathbb{Z}_3$. [S]

Quantum Echoes (October 2025)

Out-of-time-order correlators (OTOC) on Willow show harmonic propagation of quantum information through constructive interference — an echo pattern that returns periodically. FFGFT classification: In the torus formalism, modes are defined on a closed geometry; information propagation returns to the source — the echo is a consequence of compactness, not a coincidence.

11.3 The p-Bit: Between Bit and Qubit

The p-bit (probabilistic bit) occupies the intermediate range between the classical bit (deep potential well, $U \gg k_B T$, stable over years) and the qubit (metastable superposition, $U \ll k_B T$, decays in picoseconds to milliseconds). It is a magnetic nanostructure on the geometric scale $L_8 \approx 22$ nm, at which the torsion barrier becomes thermally surmountable.

Physical Basis

The energy barrier is the Stoner-Wohlfarth energy $U = K_u \cdot V$, with the magnetic anisotropy energy K_u and the volume V . The switching behavior follows the Néel-Arrhenius law: $\tau = \tau_0 \exp(U/k_B T)$. At 22 nm and room temperature, τ is in the range of microseconds to milliseconds — not a stable bit, but also not a coherent qubit.

The FFGFT classification: The switching behavior is deterministic on the sub-Planck time scale $t_0 = t_P/7500 \approx 3,6 \times 10^{-47}$ s; what appears as thermal stochasticity is the macroscopically unresolved sampling of this deterministic dynamics. The apparent randomness is epistemic, not ontological. [S]

Reconfigurable Function

By choice of the magnetic bias, the same p-bit can be operated as a stochastic switch, as a forced switch, or as an analog comparator. This makes p-bits into building blocks for probabilistic circuits (p-bit networks), which can be trained like Boltzmann machines and tackle NP-hard problems similarly to CIMs through physical relaxation.

FFGFT classification: The p-bit is not a quantum object (no coherent superposition state), but a thermally fluctuating classical mode. In the torus formalism, it corresponds to a mode whose phase θ is fully randomized by thermal noise — the limiting case of maximum decoherence on the carrier.

Why 22 nm: The Geometric Threshold

The critical size for a p-bit at room temperature corresponds to the FFGFT level $N = 8,00$, i.e., $L_g \approx 22$ nm. This is the same threshold at which classical transistors reached the quantum limit in 2012 (Chapter 10). Three independent phenomena converge on the same geometric scale: transistor limit, excitonic Bohr radius, p-bit threshold. [S]

11.4 Photon Chip: Room-Temperature Quantum Photonics

China's CHIPX/Touring-Quantum chip (2025) is a monolithic 6-inch thin-film lithium niobate (TFLN) wafer with over 1000 optical components. Key data: 1000× speedup for parallel tasks, 100× lower energy consumption, stable at room temperature, 12000 wafers per year production. [E]

Why room temperature works: Optical photons have $E \approx 1\text{--}2$ eV; $E/k_B T(300\text{ K}) \approx 40$. Thermal noise cannot create optical photons; the system sits far below the thermal threshold (Chapter 8). The TFLN single crystal with $\sim 10^{23}$ lattice sites is geometrically extremely robust. [K]

FFGFT classification: Massless modes (photons) are in the FFGFT three-orbits in $GF(27)^*$ — structurally separated from the massive fermions (fixed points $\{+1, -1\}$) (Frobenius separation, Doc. 339). [B] The photonic platform is suitable for loophole-free

Bell tests and is thus the natural testbed for the ξ deviation $\Delta\text{CHSH} \approx \xi/(2\pi) \approx 2 \times 10^{-5}$ — if detection efficiency and coherence times suffice.

Proposed FFGFT optimizations: Three approaches are conceptually sketched, all not yet realized: (1) Topology compiler that performs topological qubit placement according to the D_4 lattice structure and could reduce SWAP gates by 30–50 %; (2) harmonic resonance with qubit frequencies at ξ multiples; (3) time field modulation for active coherence preservation. Status: [S] for all three.

Sources: Doc. 083 (Photon chip 2025), Doc. 161 (Coherent Ising Machine), Doc. 183 (Google Willow, April 2026), Doc. 184 (p-Bit), Doc. 339 (Frobenius separation), Doc. 343 (Resolution floor)

Chapter 12

Limits of Quantum Computing: RSA, Weyl, and the Resolution Floor

12.1 What Quantum Computers Can and Cannot Do

This chapter draws three limits that the corpus establishes with justification. They are not speculative restrictions, but consequences of mathematical theorems or empirical findings.

12.2 Limit 1: RSA Factorization and the Arithmetic Carrier

RSA (Rivest, Shamir, Adleman 1977) relies on the difficulty of factorizing large semiprimes $N = p \cdot q$. Shor's algorithm solves the problem in polynomial time — in principle. The question is what the geometry parameter ξ can contribute.

The result of the corpus is clear: ξ **has no function in $\mathbb{Z}/N\mathbb{Z}$** . [K] The period r of the modular power function $f(x) = a^x \bmod N$ is determined by $r = \text{lcm}(p-1, q-1)$ — an individual arithmetic property of N , not a geometric parameter. Shor's algorithm uses this period structure; it is a property of the primes, not of the FFGFT geometry.

The tests in Doc. 075 and Doc. 076 use raster parameters σ with values such as $1/42$, $1/100$, $1/1000$ for period search methods. These quantities are numerical sampling widths, not physical parameters; they must not be equated with $\xi = 4/30000$. The resonance picture is in principle correct — period finding is a spectral problem — but the generators of the period are the primes, not an adjustable geometry parameter.

An advantage over classical algorithms must become visible on the Shor factorization curve: The error correction threshold must be maintained for relevant key lengths (2048 bit); the preparation must be executed in superposition; an RSA-relevant N must be successfully factored. None of this is realized as of 2026. [X]

12.3 Limit 2: Weyl Obstruction and Counting Density

The Weyl law determines how many eigenvalues a compact state space can have below a bound λ : $N(\lambda) \sim C \cdot \lambda^{d/2}$ — a power law in dimension d . [K]

The prime counting function $\pi(T)$ and the Riemann zeros grow on the other hand like $T \ln T$ — not a power, but quasi-logarithmic. This means: A finite compact state space (an n -qubit register is such a thing) cannot fully carry such a spectrum. The Weyl obstruction excludes three geometric approaches to the Riemann hypothesis: [K] no compact geometry can provide $T \ln T$ growth from a Laplace spectrum; ξ does not shift the critical line; the ξ ladder does not discretize the zeros.

Shor's circuit model is digital and not directly affected by the Weyl limit. But a physical quantum computer is a resonance system; as long as the threshold theorem for relevant scalings is not demonstrated, it lies within the resonance limits. [S]

12.4 Limit 3: The Resolution Floor

The ξ correction to Bell correlations is $\Delta\text{CHSH} \approx \xi/(2\pi) \approx 2 \times 10^{-5}$ per measurement. [S] That is the resolution floor: Below this threshold, no FFGFT-specific deviation from standard QM is measurable. All NISQ

processors (NISQ = Noisy Intermediate-Scale Quantum) operate with physical error rates $\sim 10^{-2}$, three orders of magnitude above the ξ signal.

This means for the quantum computer: The ξ corrections to algorithms — Bell damping in Shor, modified gate angles through K_{frak} — are buried under NISQ noise. [K] Neither a slowdown nor an acceleration of quantum algorithms through ξ is measurable with current hardware. This is not a failure, but a precision statement.

Future high-precision Bell tests (loophole-free, detection efficiency $> 99\%$, optical platform) could reach the limit. The photonic system (Chapter 11) is the natural candidate.

12.5 What Remains

Of the four parallel structures between FFGFT and quantum algorithms remain after this limit-setting:

The translatability (Chapter 13) is complete and symmetric. Every QM calculation is formulable in FFGFT; every FFGFT statement about quantum systems is mappable in QM. This is not a restriction, but a strength: Practitioners can keep their familiar formalism.

The added value of the FFGFT lies not in different algorithmic results, but in the origin of the input values: Masses, couplings, and α from the same geometry that carries the quantum formalism. That is the structural asymmetry that Chapter 13 has established.

Sources: Doc. 075 (RSA method), Doc. 076 (RSA period tests), Doc. 147 §8 (IBM validation), Doc. 316 (Riemann, Weyl limit), Doc. 343 (Resolution floor)

Part IV

State Space

Chapter 13

Two Languages, One Content: FFGFT and Hilbert Space

13.1 The Claim

The FFGFT describes quantum systems in the language of modes on a torus. Standard quantum mechanics describes them in the language of vectors in a Hilbert space. This chapter shows that both languages carry the same content: Every calculation that is possible in one is possible in the other, with the same result. The translation is not an approximation and not a specialization, but a bijective mapping of states, operators, gates, product spaces, and measurement prescriptions. [B]

This is methodologically the most important statement of this book. It means that the FFGFT does not replace quantum mechanics, but extends it: Both formalisms yield, up to a ξ correction of order 10^{-5} , the same measurable predictions. They differ in the interpretation of some basic concepts — Born rule, nonlocality, the status of $\hbar$, the role of space-time, singularities. Anyone at home in Hilbert space language can use the FFGFT without changing their tools. The bridge is the translation table.

13.2 Where We Are Geometrically

Before the tables come, a picture of the geometry in which the translation takes place. We will calculate with cylindrical coordinates, and the cylinder is not the full FFGFT geometry, but a convenient reduction.

The full geometry is four-dimensional. The space on which all statements of the FFGFT live is the torus $T^4 = T^3 \times S^1$: three spatial winding directions and one temporal. The wave function of a mode $\varphi_{n,l,j}$ is a function on this torus. From its geometry follow the masses, the fine structure constant, the cosmological quantities — everything that distinguishes the FFGFT from pure quantum mechanics.

A single qubit needs less. A two-level system at fixed time, without mass dynamics, does not activate most of these winding directions. Two rotational degrees of freedom remain: one that encodes where between $|0\rangle$ and $|1\rangle$ we are (this becomes z), and one for the phase (this becomes θ). The natural space for this is a 2-torus $T^2 = S^1 \times S^1$.

From the 2-torus to the cylinder. If the major radius R of the 2-torus is large compared to the tube radius, the geometry looks locally like a cylinder $\mathbb{R} \times S^1$. One of the two loops is cut open: Two circles become one line and one circle. In cylindrical coordinates (z, r, θ) , z is therefore a linear axis from -1 to $+1$, no longer cyclic.

From the cylinder to the Bloch sphere. If one pulls the cylinder together at the top and bottom into one point each, a sphere S^2 arises: the Bloch sphere of standard quantum mechanics. It is the second, further reduction.

Geometry	Dim.	Use
4-torus T^4	4	full FFGFT: masses, α , cosmology
3-torus T^3	3	mode space at fixed time
2-torus T^2	2	single qubit without reduction
Cylinder $\mathbb{R} \times S^1$	2	limit $R \rightarrow \infty$, local approximation
Bloch sphere S^2	2	poles pulled together; standard QM

What the reductions cost. Each level loses something. From the 4-torus to the 3-torus, the time evolution is lost and with it every statement about masses. From the 3-torus to the 2-torus, one spatial winding is lost and with it the connection to the third lepton generation and to many spin structures. From the 2-torus to the cylinder, one compact loop is lost — that is the weakest reduction, and exactly here sits the fractal correction factor $K_{\text{frak}} = 1 - 100\xi \approx 0,9867$, which appears in the bit-flip rotations. K_{frak} is the cylinder's memory of its torus origin.

The translation tables of this chapter apply to the qubit sector, i.e., to the bottom three rows. Standard QM works on the Bloch sphere, the FFGFT on the cylinder or the 2-torus. The translation between these levels is lossless, provided one carries the small geometric corrections (K_{frak} , ΔCHSH). The 4-torus lies one level higher and provides what the Hilbert space cannot provide: the values of the input parameters.

***For all readers:** One can draw a map in different ways: as a globe, as an unrolled cylinder, as a flat sheet. All show the same cities in the same places. But only the globe shows that one arrives back in the west from the east. The Bloch sphere of quantum mechanics is the flat sheet; the torus of the FFGFT is the globe. For navigation between two cities, the sheet suffices. For the question of why the world is as large as it is, one needs the globe.*

13.3 States

In standard QM, a qubit is a vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad \alpha, \beta \in \mathbb{C}. \quad (13.1)$$

With global phase freedom, this is a point on the Bloch sphere (Hopf fibration $S^3 \rightarrow S^2$).

In the FFGFT, the same qubit is a point (z, r, θ) on the cylinder:

$$z \in [-1, 1], \quad r \in [0, 1], \quad \theta \in [0, 2\pi), \quad z^2 + r^2 = 1. \quad (13.2)$$

z is the projection onto the computation axis, r the radial superposition amplitude, θ the phase.

The bridge between the two representations:

$$\alpha = \sqrt{\frac{1+z}{2}} e^{i\theta/2}, \quad \beta = \sqrt{\frac{1-z}{2}} e^{-i\theta/2}. \quad (13.3)$$

Conversely:

$$z = |\alpha|^2 - |\beta|^2, \quad r = 2|\alpha\beta|, \quad \theta = \arg \alpha - \arg \beta. \quad (13.4)$$

The consistency is checked in four lines: The normalization $|\alpha|^2 + |\beta|^2 = \frac{1+z}{2} + \frac{1-z}{2} = 1$ holds; the cylinder condition $z^2 + r^2 = (|\alpha|^2 - |\beta|^2)^2 + 4|\alpha|^2|\beta|^2 = 1$ holds; $|0\rangle$ corresponds to $(1, 0, *)$ and $|1\rangle$ corresponds to $(-1, 0, *)$, with arbitrary phase at the poles.

The conceptual difference lies not in the numbers. In QM, $|\alpha|^2$ is a probability. In the FFGFT, $r^2 = 1 - z^2$ is an energy density of the radial configuration. Both quantities are numerically equal for large statistics; conceptually, the FFGFT is a deterministic geometric model whose statistical predictions coincide with the Born rule.

13.4 Operators

The three Pauli matrices generate all single-qubit operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (13.5)$$

with the algebra $\sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i\epsilon_{ijk} \sigma_k$.

On the cylinder, three geometric rotations correspond to them: σ_z is the rotation about the cylinder axis, i.e., the phase rotation $\theta \rightarrow \theta + \pi$; σ_x is the rotation in the (z, r) plane by the angle π , the bit flip; σ_y is the rotation orthogonal to it, which combines bit flip and phase rotation.

Operator	Hilbert Space	FFGFT Geometry
σ_z	$\text{diag}(1, -1)$	phase rotation $\theta \rightarrow \theta + \pi$
σ_x	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	(z, r) rotation by πK_{frak}
σ_y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	rotation with phase flip
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	swap $z \leftrightarrow r, \theta \rightarrow \theta + \pi/2$
T	$\text{diag}(1, e^{i\pi/4})$	phase rotation $\theta \rightarrow \theta + \pi/4$

The only FFGFT-specific deviation is the factor $K_{\text{frak}} \approx 0,9867$ in the σ_x rotation. It follows from the fractal space-time dimension $D_f = 2,999867$ and is a testable prediction: All π gates deviate systematically by $\Delta\alpha \approx 0,013\%$ from the ideal value. [S] In the Hilbert space formalism, this would be an ad hoc correction that would have to be built in as hardware noise; in the FFGFT it follows from the geometry.

13.5 Gates

Every unitary single-qubit gate $U \in U(2)$ is a Bloch rotation:

$$U = e^{i\alpha} R_{\hat{n}}(\phi) = e^{i\alpha} \left(\cos \frac{\phi}{2} \mathbb{1} - i \sin \frac{\phi}{2} \hat{n} \cdot \vec{\sigma} \right). \quad (13.6)$$

In the FFGFT, this is the rotation of the point (z, r, θ) about the axis $\hat{n}$ by the angle ϕ . The translation is exact.

The CNOT gate

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (13.7)$$

translates as a prescription: If $z_A = -1$ (control qubit at $|1\rangle$), perform the bit flip on qubit B. This is a geometric conditioning — the rotation of one cylinder point depends on the z value of the other. Topologically, this is a linking of the two cylinders, and that is the entanglement.

The set $\{H, T, \text{CNOT}\}$ is universal for quantum mechanics according to the Solovay-Kitaev theorem. [E] Since each of these gates is representable in the FFGFT as a geometric operation, the FFGFT is fully universal for quantum computation: There is no QM statement that cannot be formulated in FFGFT language. [B]

13.6 Product Spaces and Entanglement

For n qubits, the Hilbert space is $\mathcal{H}_n = \bigotimes_{i=1}^n \mathbb{C}^2$ with dimension 2^n , and a general state reads

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \sum_x |\alpha_x|^2 = 1. \quad (13.8)$$

In the FFGFT, n qubits correspond to n mode configurations (z_i, r_i, θ_i) on a common torus geometry. Entanglement is a topological connection: Entangled qubits are knots of the mode configuration that cannot be reduced to product form. The Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is a topologically connected cylinder configuration in which the individual (z_i, r_i, θ_i) are not independent.

Both formalisms reproduce the exponential growth $\dim = 2^n$: in the Hilbert space through the tensor product, in the FFGFT through n independent mode configurations with topological connections. The number of real parameters per state is $2 \cdot 2^n - 2$ in both cases.

The CHSH prediction: Standard QM gives $|\langle \text{CHSH} \rangle| \leq 2\sqrt{2}$. The FFGFT predicts a small, geometrically motivated deviation, $\Delta \text{CHSH} \approx \xi/(2\pi) \approx 10^{-5}$ per measurement. [S] It lies below the current hardware noise threshold; the distinction from the cumulative quantity from Doc. 022 is explained in Chapter 7.

13.7 Measurement

A measurement in the computational basis yields in QM the value 0 with probability $|\alpha|^2$ and the value 1 with probability $|\beta|^2$; the state collapses to $|0\rangle$ or $|1\rangle$.

In the FFGFT, measurement is not a probabilistic collapse, but a geometric interaction that drives the point (z, r, θ) to the nearest stable extremum $z = \pm 1$. This is a deterministic movement in mode space — because of the unknown phase θ and smallest variations of the initial conditions, statistically indistinguishable from the behavior of the Born rule. For large statistics, the measurement distribution converges:

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0). \quad (13.9)$$

The formalisms yield identical statistical predictions. [B] The dispute over local realism resolves in the FFGFT geometrically, because entanglement is represented as a topological connection and not as a nonlocal correlation (Chapter 5).

13.8 Time Evolution

The time evolution of QM is the Schrödinger equation $i\hbar \partial_t |\psi\rangle = H|\psi\rangle$. In the FFGFT, it is the deterministic movement of the mode configuration $\varphi_{n,l,j}$ on $T^3 \times \mathbb{R}$. The mode equation reduces in the low-energy limit $E \ll E_\xi$ to the Schrödinger equation:

$$i\hbar \frac{\partial \varphi_{n,l,j}}{\partial t} \xrightarrow{E \ll E_\xi} -\frac{\hbar^2}{2m} \nabla^2 \varphi_{n,l,j} + V \varphi_{n,l,j}. \quad (13.10)$$

At higher energies, ξ corrections occur, which in the Hilbert space formalism appear as modifications of the Hamilton operator.

13.9 What Is Translatable and What Is Not

All Hilbert space statements are mappable in FFGFT: States, operators and gates (with or without K_{frak}), tensor products, entanglement, Bell and GHZ states, measurement, unitary evolution and all algorithms. Conversely, every (z, r, θ) configuration corresponds

uniquely to an (α, β) vector, every geometric rotation to a unitary operator, every topological connection to an entangled vector. Every calculation that is performable in one formalism is performable in the other — with identical result.

The translatability holds for statements about quantum systems. It does not hold for statements about the input values of quantum mechanics:

- The parameter $\xi = 4/30000$ is an external input value in the Hilbert space. In the FFGFT it follows from the geometry (Chapter 1).
- The fermion masses appear in the Hilbert space as Yukawa couplings. In the FFGFT they follow from $m_i = r_i \xi^{p_i} v$.
- The fine structure constant is derivable in the FFGFT, input in the Hilbert space.
- K_{frak} appears in FFGFT gates as geometric deviation; in the Hilbert space it would have to be modeled as noise.

That is the structural asymmetry: The Hilbert space has all tools for QM calculations, but no explanation for the input values. The FFGFT provides the tools and the input values from a single geometry. In this sense, quantum mechanics is a subset of the FFGFT: the FFGFT, restricted to the question of how systems behave, without the question of why the constants have these values.

The FFGFT does not claim to be the only correct representation. It is a representation that additionally provides the values of the fundamental constants. Anyone who does not need this added value can ignore it; standard QM remains valid.

13.10 Operational Equivalence, Interpretive Divergence

The complete translatability must not be read as ontological identity. Operationally, FFGFT and standard QM agree in every measurable result up to ξ corrections. Interpretively, they diverge at five points where the standard reading makes statements that are read in the FFGFT as artifacts of the chosen carrier assumptions:

1. **Born rule as irreducible chance.** Standard reading: $|\psi|^2$ is ontological probability, the measurement outcome fundamentally random. FFGFT: $|\psi|^2$ is the correct distribution of observations,

but arises from a deterministic pole transition dynamics. The stochasticity sits in the measuring apparatus, not in the natural process.

2. **Nonlocality as a property of nature.** Standard reading of the Bell correlations: action at a distance in $\mathbb{R}^3$. FFGFT: topological connection on T^4 , geometric proximity in a closed geometry.
3. **$\hbar$ as primitive.** Standard reading: fundamental constant of nature. FFGFT: SI conversion factor between energy and time measure, which follows from the closure condition $S = h$ on T^4 . Numerically identical, ontologically derived.
4. **Space-time as stage.** Standard reading of the theory of relativity: given manifold. FFGFT: derived from the T^4 identification geometry. Questions like “what was before the Big Bang” are artifacts of the $\mathbb{R}^3 + t$ assumption.
5. **Singularities as realities.** Standard reading: Black hole and Big Bang as true singularities. FFGFT: Since T^4 is closed, they are endpoints of the carrier model, not of the physics. Outside the singularity region, the predictions agree.

These five points are not empirical claims against standard QM. They are interpretation differences that become visible as soon as one asks about the ontological carrier. Anyone who only calculates with the numbers can ignore them. Anyone who wants to know why the numbers are these specific ones will find them to be testable structural statements.

Within today’s measurement precision — NISQ level $\sim 10^{-2}$, optical precision experiments $\sim 10^{-12}$ in Bell tests — both yield the same predictions. The ξ correction $\sim 10^{-5}$ is not resolvable with current hardware; this has been empirically tested on IBM Heron r2 (Chapter 7).

13.11 Field-Theoretic Extension

The translation can be extended from the qubit sector to quantum field states. The Fock space of standard QFT — the direct sum of all n -particle spaces — corresponds to the mode spectrum of the carrier, where the occupation numbers appear as winding numbers.

Creation and annihilation operators are transitions between neighboring winding states of the same mode; commutators follow from the periodicity. This generates no new predictions in the perturbative regime, but restructures the construction of the Standard Model geometrically: The particle species are mode classes of the lattice, not independent fields with their own coupling each. [S]

What This Chapter Holds

States, operators, gates, entanglement, measurement, and time evolution are bijectively translatable between FFGFT and Hilbert space. The only formal deviation is K_{frak} in the π gates; the only statistical one is ΔCHSH . The added value of the FFGFT lies not in different calculation results, but in the fact that the input values of quantum mechanics follow from the same geometry.

Sources: Doc. 230 (Translatability, May 2026), Doc. 231 (field-theoretic extension), Doc. 147 §1.2, Doc. 175 §3–4, Doc. 202 §22

Chapter 14

The Lattice in Hilbert Space

14.1 Do the Lattices Become Flat?

The FFGFT calculates geometrically on T^4 with the D_4 lattice and quantum mechanically on $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$. During the transition, an obvious question arises: Do the lattices become flat? Does anything remain of kissing number 24, packing density, and triality, or is every lattice the same in the Hilbert space?

The answer: The torus is flat anyway — curvature was never the difference —, and the lattice structure is not lost, but **migrates into the spectrum and into the fiber**. In the Hilbert space, the lattice becomes the index set of the Fourier modes. Thus it is no longer a sphere packing; in this sense, every lattice becomes flat. But the information does not disappear: The degeneracies of the Laplace operator are the theta series of the lattice, the kissing number becomes the degeneracy of the first excitation, the triality becomes the orbifold and fiber structure.

$$\boxed{\text{Geometry} \rightarrow \text{Spectrum} + \text{Fiber}} \quad (14.1)$$

Two anchors keep this translation stable. First, D_4 is self-dual up to scaling: The dual lattice D_4^* is again a D_4 lattice, the Fourier transform does not lead out of the family. Second, the fiber changes nothing about the lattice: The $\mathbb{C}^3$ fiber hangs at every point and carries the $\mathbb{Z}_3$ structure, the mode lattice remains the dual lattice.

On notation. In lattice theory, Λ is the usual symbol for a lattice. In the FFGFT corpus, Λ is occupied (cosmological constant, closure scale). This chapter therefore uses Γ for the lattice and Γ^* for the

dual lattice; the edge lengths of the torus are called R_{123} (three spatial directions) and R_4 (fourth direction), their ratio $r = R_4/R_{123}$.

14.2 The Spectrum Separates What Curvature Does Not

The complete shell counting up to $|k|^2 = 12$, for the cubic lattice $\mathbb{Z}^4$ and for D_4 (all integer k with even coordinate sum):

Norm $ k ^2$	$\mathbb{Z}^4$	D_4	D_4 -Split ($k_4=0$ / $k_4 \neq 0$)
1	8	0	—
2	24	24	12 / 12
3	32	0	—
4	24	24	6 / 18
5	48	0	—
6	96	96	24 / 72
7	64	0	—
8	24	24	12 / 12
9	104	0	—
10	144	144	24 / 120
11	96	0	—
12	96	96	8 / 88

Three readings. **Two flat tori are spectrally distinguishable:** $\mathbb{Z}^4$ begins at norm 1 with degeneracy 8, D_4 at norm 2 with degeneracy 24. For D_4 , the odd norms are completely absent — the parity condition excludes them. What curvature does not distinguish, the spectrum distinguishes immediately. **[K] Independent control:** The D_4 series 24, 24, 96, 24, 144, 96, ... is the known theta series $\theta_{D_4} = (\theta_3^4 + \theta_4^4)/2$; the script calculation reproduces it coefficient by coefficient. **The 12 + 12 decomposition is spectrally addressable:** The first excitation carries exactly twelve modes with $k_4 = 0$ (the spatial FCC half) and twelve with $k_4 \neq 0$ (the winding half). At higher shells, the splitting is asymmetric — 6/18 at norm 4, 8/88 at norm 12. The 12/12 symmetry is a property of the first shell, not a general statement; where the corpus argues with $24 = 12 + 12$, this holds exactly where it is done.

14.3 Anisotropy: Where $\tilde{T} \cdot m = 1$ Becomes Readable in the Spectrum

If the fourth direction is compactified with a different radius, the eigenvalue is $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$, and the first shell splits:

r	E (winding, 12 modes)	E (FCC, 12 modes)
1,0	2,000	2,000 (degenerate: 24)
1,1	1,826	2,000
1,5	1,444	2,000
2,0	1,250	2,000

The FCC modes remain fixed at $E = 2$, the winding modes follow exactly $E_{\text{winding}} = 1 + 1/r^2$. The splitting is a direct measure of the radius ratio — the place where the time-mass structure becomes readable in the spectrum: Mass direction and spatial directions carry different excitation energies as soon as they are of different lengths. [K]

All levels have the form $E = a + b/r^2$ with integer a, b . In the range $1 < r \leq 2,5$, among all levels up to norm 8 there are exactly two crossing radii: At $r = \sqrt{2}$, the pure winding $(0, 0, 0, \pm 2)$ crosses the FCC twelve; at $r = \sqrt{3}$, the winding twelve. The level sequence thus does not change continuously, but at two distinguished root radii; every ordering statement about the lowest excitations decomposes into exactly three regimes. Beyond $\sqrt{3}$, the fundamental tone of the spectrum is a pure winding.

14.4 The Reciprocity Is Exact Mode-by-Mode — as a Position Relation

The fourth direction is the time-mass dual; it is unrolled onto time. Two test questions: How do time periods arise, and does $\tilde{T} \cdot m = 1$ hold only on average?

The unrolling of the compact fourth direction is the universal covering $\mathbb{R} \rightarrow S^1$; the deck transformation group is $\mathbb{Z}$, the unrolled coordinate counts cycle traversals. A k_4 mode thereby becomes a periodic function with period $2\pi R_4/k_4$. Since $\pi_1(S^1) = \mathbb{Z}$ is abelian, this bookkeeping is lossless.

In the deformation spectrum, $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$ is literally $E^2 = p_{\text{space}}^2 + m^2$ with $m_k = k_4/R_4$: **The winding number is the mass number.** Thus

$$\lambda_4 \cdot m = \frac{2\pi R_4}{k_4} \cdot \frac{k_4}{R_4} = 2\pi \quad \text{exact, for every mode.} \quad (14.2)$$

The reciprocity holds not only on average — it is identically fulfilled mode-by-mode, because period and mass are built from the same quantum number and the same radius. [K]

But the period is initially a position, not a duration. The quantity $\lambda_4 = 2\pi R_4/k_4$ is an arc length on the circle: the distance in x_4 after which the mode repeats. The relation $\lambda_4 \cdot m = 2\pi$ is the de Broglie relation on the mass direction — wavelength times wavenumber —, not a time statement. Duration arises only from the unrolled coordinate: A point of the unrolled time is a pair $t = w \cdot 2\pi R_4 + x_4$ of revolution number $w \in \mathbb{Z}$ and position on the cycle. The duration is entirely in the numerator w , the position in the remainder. On the cycle there is no distinguished beginning; every moment is a place that recurs.

The temporal reading carries a γ factor. If one asks about the oscillation period in the unrolled time, then $T_{\text{osc}} = 2\pi/E$ with $E = \sqrt{p^2 + m^2}$ is responsible, and then

$$T_{\text{osc}} \cdot m = \frac{2\pi}{\gamma} \leq 2\pi, \quad \text{equality exactly at } p_{\text{space}} = 0. \quad (14.3)$$

Exactly 2π is the position reading; the time reading coincides with it only for the resting mode.

In the mixture, the equation becomes an inequality. If a state superposes several k_4 shells with weights p_k , then $\langle T \rangle \langle m \rangle = 2\pi \langle k \rangle \langle 1/k \rangle$ and by Jensen $\langle k \rangle \langle 1/k \rangle \geq 1$, with equality exactly when a single shell is occupied. Numerically: sharp shell exactly 1; $k = 1, 2$ equally occupied 1,125; uniform distribution $k = 1 \dots 50$ yields 2,295; thermal distributions between 1,00002 and 2,6; two thousand random distributions never below 1. The conjecture reverses: The condition for $\tilde{T} \cdot m = 1$ is not uniform distribution, but **sharpness** — exactly one occupied mass shell. The excess measures the mass dispersion of the state.

Thus three statements stand in three directions: The position reading is always 2π ; the time reading of a moving mode is $2\pi/\gamma$; the mass mixture lies at $\geq 2\pi$. The exact relation holds for the resting,

sharp mode — there and only there do both correction factors vanish simultaneously. Motion pushes the product down, mass dispersion raises it. This is not a contradiction to the native FFGFT formulation, but the Kaluza-Klein reading of its validity conditions.

***For all readers:** A guitar string has a fundamental frequency. If one strings it and plucks it, it vibrates at exactly this frequency — the relation “length times frequency” is exact. But if one strikes several strings at once, or if the guitar moves past one, one hears something else: a chord, or a Doppler tone. The FFGFT relation $\tilde{T} \cdot m = 1$ is the string at rest; everything else are corrections that can be specified exactly.*

14.5 Rolled Up and Unrolled: The Closure Bifurcation Is Spectrally Invisible

The corpus poses the closure question as a bifurcation: The defect $d = 100\xi = 1/75$ per revolution allows three cases for the unrolled time. With the cycle length $\tau_c = 2\pi/H_0 = 91,9$ Gyr: Case A without defect closes after one revolution; Case B with frozen ξ after 75 revolutions (rational rotation number $74/75$, coprime), thus after 6892 Gyr; Case C with running ξ never — the winding becomes a logarithmic spiral.

These numbers are deadlines, not repetitions. They do not say that something repeats, but within what deadline a recurrence would have to take place if the respective case were true. The geometry is closed; whether history returns is independent of that. The “Big Bang” is, in this reading, a place on the cycle — the region of smallest masses and slowest clocks —, not an event; a closed circle has no starting point.

It would be obvious to read the defect as a twist of the boundary condition ($k_4 \rightarrow k_4 + 1/75$). That is wrong, and the reason is structural: **In the rolled-up state, there are no fractal corrections.** The uniqueness of the mode on the circle enforces $k_4 \in \mathbb{Z}$ — integer or not at all. A winding number is topological; it has no place for $1/75$. Corrections need something that accumulates, and only a traversed path can accumulate. Already the formulation “defect per revolution” presupposes the counting of revolutions, and revolutions are only counted in the unrolled coordinate.

Quantity	State	Fractal Correction
Theta series, degeneracies, $24 = 12 + 12$	rolled up	no
$\text{Aut}(D_4)$, triality, orbifold	rolled up	no
Circulant, container 4×6	rolled up	no
Casimir, heat kernel, congruences	rolled up	no
$\lambda_4 \cdot m = 2\pi$	rolled up	no
Cycle length, defect, Case A/B/C	unrolled	yes
Revolution number w , history	unrolled	yes
T_{osc} , γ factor	unrolled	yes

Two consequences. First, the bifurcation has no spectral signature: The mode spectrum is an object of the rolled-up state; whether the path closes after one, after 75, or after no revolution changes nothing about the eigenvalues. Second, all following sections are case-independent: Orbifold, circulant, and container are statements about the compact geometry, and there the defect does not apply. [K]

14.6 Reorganizations: Three Levels and One Limit

Flat lattices can be reorganized on three levels that cost very different things. **Basis change** (free): Every unimodular transformation from $GL(4, \mathbb{Z})$ describes the same point lattice with different generators; the spectrum changes by nothing. **Lattice symmetries** (free, structure-bearing): Reorganizations that map the lattice onto itself — here sits the central finding of the next section. **Rebuilding to a different lattice** (costs physics): discretely via glue vectors along the chain $D_4 \subset \mathbb{Z}^4 \subset D_4^*$ with index 2 each, or continuously through deformation in the moduli space. Each step shifts degeneracies; D_4 is a distinguished point therein — kissing number 24, provably unsurpassable in four dimensions.

The limit of the spectral statement: From dimension 4 onward, pairs of flat tori exist that are isospectral but not isometric (Schie-mann). The general statement “spectrum determines the lattice” is

false in 4D. For D_4 itself uncritical — the kissing number 24 distinguishes it uniquely — but as a rule it must not be booked.

14.7 Triality Is Built In, Not Attached

Calculated on the lattice, by brute-force counting of all basis tuples with identical Gram matrix:

$ \text{Aut}(D_4) $	$1152 = W(F_4) $
$ \text{Aut}(\mathbb{Z}^4) $	$384 = 2^4 \cdot 4!$
$ \text{Aut}(D_4) / W(D_4) $	$1152/192 = 6 = S_3 $

The quotient $6 = |S_3|$ demonstrates triality directly at the lattice automorphism group, not only at the Dynkin diagram: D_4 is the only lattice of the D_n family with three equal Dynkin arms; for $D_5, D_6, \dots$ the outer symmetry shrinks to $\mathbb{Z}_2$. [K]

The 80 elements of order 3 decompose into three classes. Calculation is exact: The triality elements are half-integer — the glue vector mixture $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ enters; rounding to integers destroys them, therefore one calculates with $2A$.

Class	Number	Character
Trace -2 , half-integer	16	fixed-point free in $\mathbb{R}^4$; eigenvalues $\{\omega, \bar{\omega}\}$ doubly
Trace 1, integer	32	fixed plane, free on the 24 minimal vectors
Trace 1, half-integer	32	6 fixed vectors + 6 three-orbits

The orbifold is lattice symmetry, not an additional assumption.

For each of the 16 elements with trace -2 , exactly $|\det(1 - A)| = 9$; the quotient $T^4/\langle A \rangle$ is a $\mathbb{Z}_3$ orbifold with exactly nine fixed points, all in third coordinates. The $T^4/\mathbb{Z}_3$ structure does not need to be postulated; it is already present in D_4 . [K]

On every mode orbit, the $\mathbb{Z}_3$ acts as a circulant. The unitary action on $L^2(T^4)$ permutes the three plane waves of a three-orbit cyclically; the permutation matrix has eigenvalues $\{1, \omega, \omega^2\}$, phases $0^\circ, +120^\circ, -120^\circ$ — structurally exactly the circulant diagonalization of the $\mathbb{C}^3$ fiber. The first shell yields eight copies: $24 = 8 \times 3 = 8 \times (\chi_0 \oplus \chi_1 \oplus \chi_2)$.

14.8 The Negative Finding: 5 Does Not Divide 1152

The order distribution in $\text{Aut}(D_4)$, determined exactly:

Order	1	2	3	4	6	8	12
Number	1	139	80	228	464	144	96

Sum 1152. **Order 5: zero elements** — not a calculation result, but a theorem: $1152 = 2^7 \cdot 3^2$, and 5 does not divide that (Lagrange). [K]

Consequence for the Koide phase $\theta = 2/9$: The fivefold rotation from which $p_0 = 2/9$ follows is in principle incompatible with the D_4 lattice structure. The test was additionally conducted directly: All invariants of the linear lattice action at the nine orbifold fixed points have the denominator spectrum $\{1, 2, 3, 6\}$ — no ninths. This does not devalue the derivation of $2/9$, but delimits it: The phase belongs to a layer *above* the lattice (the A_5 structure), the lattice carries the $\mathbb{Z}_3$. The division of labor is thus exactly named.

14.9 The Doubling Is Nevertheless Prepared

Does the negative finding contradict the doubling $6 = 3 \oplus 3'$? No, in three stages.

The prohibition hits the wrong level. $5 \nmid 1152$ forbids order-5 lattice automorphisms. The doubling lives on the fiber ($\mathbb{C}^3 \rightarrow \mathbb{C}^6$); A_5 is nowhere demanded as a spatial symmetry. The compatibility condition is solely that the interface of both levels exists: Element orders in A_5 are $\{1, 2, 3, 5\}$, in $\text{Aut}(D_4)$ they are $\{1, 2, 3, 4, 6, 8, 12\}$; the intersection is $\{1, 2, 3\}$, canonically thus C_3 . Exactly that is provided by the triality. The interface is not only allowed, it is the only possible one — and it is occupied.

The lattice prepares the doubling. The central involution $-1 \in \text{Aut}(D_4)$ commutes with the triality $\mathbb{Z}_3$ and pairs the eight three-orbits of the first shell into four disjoint pairs. No orbit is self-paired, provable: $-k = A^j k$ would require the eigenvalue -1 for an element of order 3, impossible. Thus

$$24 = 8 \times 3 = 4 \times 6 \quad (\text{orbit} \oplus \text{mirror orbit}). \quad (14.4)$$

Each six-block carries under C_3 the character content $2 \times (\chi_0 \oplus \chi_1 \oplus \chi_2)$ — exactly the restriction of $3 \oplus 3'$. The first excitation shell yields four ready containers. [K]

The limit, cleanly booked. The labeling — which factor 3 and which $3'$ — is lattice-blind. The A_5 characters of the two triplets differ only on the 5-classes (φ versus $1 - \varphi$); on the 3-class both are zero. The distinction depends exactly on the structure that does not exist in the lattice. Balance: The lattice provides containers and C_3 content, the A_5 layer provides the labels, φ and the fivefold — and thus $2/9$.

14.10 Anchorless Results

All following quantities are dimensionless — ratios, counts, congruences. No SI scale enters.

Two congruence theorems. All complete shells up to norm 120 checked: $N(n) \equiv 0 \pmod{3}$ and $N(n) \equiv 0 \pmod{6}$ without exception. Both are theorems, not empirical: The triality $\mathbb{Z}_3$ is fixed-point free on $\Gamma \setminus \{0\}$, so every shell decomposes into three-orbits; together with the -1 pairing without self-pairing into six-blocks. [K]

Character-resolved spectrum. Consequence: exact equal distribution on every shell. The invariant sector of the orbifold has exactly $N(n)/3$ states per shell; the $\mathbb{Z}_3$ projection costs no asymmetry on any shell. Honest consequence: Equal distribution and six-containers are not a distinction of the first shell, but a theorem for all. **The structure is everywhere — and therefore can select nothing.** The pattern is present scale-invariantly; which shell is physically occupied is decided by something else.

Casimir: The densest lattice loses. The ζ -regularized vacuum energy of a scalar field, both tori normalized to covolume 1: $\mathbb{Z}^4$ torus $E = -0,9321$, D_4 torus $E = -0,8686$. The $\mathbb{Z}^4$ torus lies deeper. The reason is the spectral gap: The densest lattice has the largest first eigenfrequency at the same volume ($\sqrt{2}$ versus 1) and thus the less negative zero-point energy. Remarkable is the order reversal: For $s > 0$, D_4 minimizes the Epstein zeta, at $s = 0$ all lattices are equal, at $s = -1/2$ the order is inverted. Interpretation limit: This holds for the scalar, periodic field; "nature chooses $\mathbb{Z}^4$ " does not follow from this.

Heat kernel: Quantifying the observability limit. For both tori, $t^2\theta(t) \rightarrow 1$ (identical Weyl term at the same volume); the lattice difference is exponentially small: relative difference $1,2 \times 10^{-2}$ at $t = 0,5$, $1,8 \times 10^{-13}$ at $t = 0,1$, 4×10^{-27} at $t = 0,05$. Every observable that relies only on the leading heat kernel coefficient is lattice-blind.

14.11 What Is Still Naked Here — and What Is Not

This chapter calculates with the smooth Laplace operator. To be clarified is what the FFGFT counters with in terms of fractal structure, and here two quantities are to be cleanly separated: $D_f^{\text{space}} = 3 - \xi$ is local and acts with $6,7 \times 10^{-5}$; $K_{\text{frak}} = 1 - 100\xi$ is cumulative over the RG flow from the Planck to the electroweak scale (17 orders of magnitude) and acts with $1,33 \times 10^{-2}$. The ratio of the two effects is exactly 200. The fractal dimension of the FFGFT is the spatial one, not a four-dimensional one; a quantity $D_f = 4 - \xi$ is not known to the corpus.

A Laplace operator on the compact torus is local. A cumulative quantity cannot be carried by it; K_{frak} does not enter via the operator, but via the bridge constants during the scale conversion. The consequence decomposes into three unequal parts.

Untouched: everything anchorless. K_{frak} applies to absolute quantities and cancels in ratios of the same level. Thus untouched remain: all counts (24, 8, $N(n)$, 8×3 , 4×6), the congruences, the character distribution, all ratios E/E_{min} and the crossing radii. These statements are fractal-invariant.

One factor: all absolute energies. With an anchor, $E = \hbar H_0 |k|$. These values carry the factor: first D_4 shell $3,230 \times 10^{-52}$ J naked, $3,187 \times 10^{-52}$ J with K_{frak} ; throughout $-1,33\%$. The factor sits not in the spectrum, but in the anchor: The dimensionless eigenvalues remain unchanged, what is corrected is the conversion to SI.

The operator, not the conversion: perturbation theory. At $D_f^{\text{space}} = 3 - \xi$, the operator is no longer an exactly smooth Laplace, and exact symmetry is precisely what carries the degeneracies. Decisive is solely the local effect, $\Delta E/E \approx 6,7 \times 10^{-5}$ — two orders of magnitude below what an estimate via K_{frak} would yield. The 24-fold degeneracy would then be a multiplet with fine structure. Into which submultiplets it decays is decided by the symmetry of

the perturbation, namely via representation theory, not the orbit structure:

Perturbation respects	Multiplet structure of the 24
full $\text{Aut}(D_4)$	$9 + 8 + 4 + 2 + 1$
triality and -1	$8 \times 1 + 8 \times 2$
only $\mathbb{Z}_3$	$8 \times 1 + 8 \times 2$
only -1	24×1
none of it	24×1

The obvious expectation is wrong. One suspects the orbits remain together. In fact, the multiplet sizes follow the irreducible representations of the respected symmetry group: Every position-dependent perturbation breaks translation invariance, and that carried the 24-fold degeneracy, not the point symmetry; $\mathbb{Z}_3$ is abelian, all irreps one-dimensional, the doublets are the antiunitary pairing of the conjugate characters; -1 alone enforces nothing; even the maximally symmetric perturbator does not keep the 24 together. Qualitatively, of the degeneracy only as much is stable as the perturbation symmetry yields in irrep dimension; quantitatively, the perturbation lies at 10^{-5} , a fine structure far below any readability, and the container structure remains practically intact. [K]

Balance: Naked are only the SI absolute values, namely via the anchor, not via the operator. Degeneracies, ratios, congruences, containers, Casimir and heat kernel ratios are not naked, but correct: K_{frak} does not structurally apply there.

14.12 The Spectral Operator $\hat{F}$

What has been calculated on the lattice thus far receives its rigorous Hilbert space embedding through the spectral operator. In linear algebra, a symmetric matrix has real eigenvalues and an orthonormal basis of eigenvectors. In infinite-dimensional Hilbert spaces, spectral theory generalizes this: For a self-adjoint operator, a unique spectral measure exists with $T = \int \lambda dE(\lambda)$, and self-adjointness guarantees a real spectrum — the condition for observables to have measurable values. [E]

The FFGFT posits a fractal operator

$$\hat{F} \equiv \bigoplus_{n=1}^{\infty} S_n \circ L_n \quad (14.5)$$

with scale operators $S_n = r_n U_n$ (contraction rates $r_n \in (0, 1)$, unitary phase rotations U_n) and logical projectors L_n with $L_n L_m = L_{\min(n, m)}$ — the fractal inclusion structure: Projections onto finer scales contain those onto coarser ones. [SETZUNG] Since the fundamental length $L_0 = \xi \ell_P$ makes the scale ladder finite, $\sum r_n < \infty$; $\hat{F}$ is bounded on all of $\mathcal{H}$, the symmetry follows from the $\mathbb{Z}_3$ pairing $(k, -k)$, the defect indices are $(0, 0)$: Exactly one self-adjoint realization exists. [B] Thus the question left open in Doc. 282 (register bridge R82) is closed; on the compact carrier: bounded and symmetric implies self-adjoint.

The eigenvalue equation $\hat{F}\psi_\lambda = \lambda\psi_\lambda$ defines the resonance modes. The geometry parameter is not a free parameter, but a spectral value:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}, \quad (14.6)$$

the smallest eigenvalue of the D_4 suboperator — the fundamental frequency of the fractal space. This structurally explains why ξ appears as a basis in all mass formulas: It is the lowest mode of the geometry operator. [K]

The state space has tensor product structure $\mathcal{H} = \mathcal{H}_{\text{geom}} \otimes \mathcal{H}_{\text{spin}} \otimes \mathcal{H}_{\text{flavor}}$ with $\mathcal{H}_{\text{geom}} = L^2(T^4/\mathbb{Z}_3, d\mu_f)$, $\mathcal{H}_{\text{spin}} = \mathbb{C}^6$ and $\mathcal{H}_{\text{flavor}} = \ell^2(\mathbb{N}^2)$ with winding numbers (n_θ, n_ϕ) as quantum numbers. The fractal measure $d\mu_f$ arises through successive fractalization of the Lebesgue measure in 100 steps with running $\xi^{(k)} = \xi(1 - k/100)$; after the recursion, the coefficients $n^\xi \approx 1 + \xi \ln n$ are nearly 1, and the fractal dimension $D_f = 3 - \xi$ emerges from the recursion process, not from a postulate.

The $\mathbb{Z}_3$ orbifold projection $P_{\mathbb{Z}_3} = \frac{1}{3} \sum_k \omega^k \tau^k$ cuts out the invariant subspace from $L^2(T^4)$. At the fixed points, the boundary condition $\psi_\chi(x_0 + y) = \chi \cdot \psi_\chi(x_0 + g_*(y))$ holds with the character χ and the linearized action g_* . The three neutrino flavors are localized at three of these fixed points; the different positions generate different eigenvalues of the mass operator. [K]

What This Chapter Holds

In the Hilbert space, the lattice becomes flat, but not without consequence: What the geometry carried, the spectrum and fiber carry. Triality is lattice symmetry; the $\mathbb{Z}_3$ orbifold with nine fixed points is contained in D_4 ; fivefold symmetry is impossible at the lattice and belongs to the A_5 fiber layer; $\tilde{T} \cdot m = 1$ is exact mode-by-mode as a position relation and holds temporally only for the resting, sharp mode. Rolled-up quantities carry no fractal correction; K_{frak} acts solely via the scale bridge. $\hat{F}$ is self-adjoint, ξ its smallest D_4 eigenvalue.

Sources: Doc. 314 (Lattice in Hilbert space, 2026), Doc. 322 (Spectral theory), Doc. 327 (Self-adjointness, R86), Doc. 133 (K_{frak}), Doc. 285/291/293/311/313

Chapter 15

Where Time Sits: State Space and Superposition

15.1 The Bifurcation

At the beginning of every quantum mechanical model building stands a choice that usually remains unspoken and nevertheless determines everything else: Is time a direction *in* state space, or does it run *alongside* it, as an external parameter?

The question sounds technical, but is ontological. It does not decide which mathematics is used — both answers work with self-adjoint operators, spectra, and projectors —, but *where* time sits. And precisely this placement decides whether the physical quantities *fall* out of the structure or must be *attached* to it.

The question reaches back to the roots of quantum theory. As early as 1932, Wolfgang Pauli recognized that a time operator canonically conjugate to the Hamiltonian encounters fundamental difficulties. The Pauli theorem has shaped the development of quantum mechanics and seemingly made time as an external parameter irrevocable. Later works — Aharonov and Bohm, Garrison and Wong — showed that under certain conditions, such as for non-bounded-below Hamiltonians or compact configurations, time operators can nevertheless be constructed. This exception is, for the FFGFT, the rule.

15.2 The Unrolled View: Time as External Parameter

The familiar formulation situates time outside the state space. The Hilbert space is the space of states *at one time*, typically $\mathcal{H} = L^2(\text{configuration space})$. Time is the parameter t along which the state evolves, mediated by $e^{-i\hat{H}t}$. Position, momentum, spin are operators on $\mathcal{H}$. Time is none.

That it cannot be one is not an oversight, but a result: Pauli's objection shows that for a Hamiltonian bounded from below, no self-adjoint, canonically conjugate time operator exists. [E] In the unrolled view, time therefore necessarily remains the stage, not the play.

The consequence is structural: Since time stands outside, the state space alone contains no dynamical scales. Masses, couplings, lifetimes are not present in the pure spatial structure — they must be introduced from outside. In the Standard Model, this happens through Yukawa terms and the free parameters that fix their strength. The structure provides the framework; the numbers come from measurement.

The problem is old. Weyl and Wigner recognized that the masses of elementary particles appear group-theoretically as Casimir invariants — as structural labels, not as attached parameters. Nevertheless, the origin of the mass values remained open. The Higgs mechanism explains the mass of the gauge bosons, but introduces new free parameters itself. The criticism of the unrolled view is not that it is wrong, but that it leaves the origin of the mass scales unanswered and refers to external inputs.

A Representative Construction of This Kind

Many model programs that strive for a geometric origin of particle properties follow, despite all differences, the same scheme: One chooses a discrete carrier (a lattice, a cut-and-project structure, a periodic graph); defines on it a self-adjoint operator family; extracts sectors via spectral projectors; and assigns physical meaning to these sectors afterwards — charge, spin, mass, the latter usually via a parametric mapping with one or two free parameters.

Such a procedure can be mathematically fully controlled. The decisive observation concerns not its rigor, but its placement of time: The carrier is a spatial object, the operator evolution runs over an external parameter. The sectors are spatial eigenspaces, not time modes. Mass is not contained in them; it is attached.

And here an argument applies that carries itself: **A parametric mapping with free parameters can hit any finite list of values.** If the mapping is not fixed in advance and universally, it possesses no predictive content — it renames the known inputs. To deliver numbers, such a construction must be coupled to external values: to measurement, and thus indirectly to the Standard Model itself. That is honest, provided it declares its limit: that it does not derive masses, but provides a hypothesis architecture. The difficulty of this path is not accidental, but the consequence of the unrolled placement of time.

A concrete example: Graph modeling defines quasiparticles on hexagonal lattices. The excitations show linear dispersion and can be read as massless fermions — but the introduction of a mass always requires additional terms (Haldane term, Kekulé distortion) with free parameters. The geometry provides the kinematics, not the mass scale.

15.3 The Rolled-Up View: Time as Compact Dimension

The alternative does not leave time outside and does not eliminate it either, but pulls it *into* the state space — as a closed, periodic dimension. In place of $L^2(\text{space})$ with external t comes a space in which a compact time direction is part of the geometry: $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$, where one of the torus directions carries time. The bijectivity of this representation (Type III) is documented in the corpus (Chapter 13).

On a closed dimension, the admissible states are standing modes with discrete winding numbers. Quantization is then not postulated, but geometrically enforced: Only certain modes fit on the closed structure. Mass becomes an eigenvalue of an operator on $\mathcal{H}$ itself — not attached, but structurally contained. The ratios of the masses are fixed, because the admissible modes are fixed. [B]

The relation $\tilde{T} \cdot m = 1$ is the concrete execution of this idea: Every massive quantity *is* its own time cycle. The decompactification $S_m^1 \rightarrow \mathbb{R}_t$ is measure-preserving, thus lossless. That a compact dimension generates a discrete mode tower whose masses behave as the inverse of the compactification period is not an exotic feature, but the mechanism that Kaluza and Klein demonstrated for a compact spatial dimension — here applied to time.

Pauli's objection does not hit the rolled-up view. It applies to an open time unbounded below conjugate to a bounded Hamiltonian. A compact time behaves like an angle variable: Its conjugate spectrum is discrete, as with angular momentum to the angle, and the Pauli argument does not apply. [B] The closed time is not only admissible, it is the natural place of a discrete mode structure.

This insight has roots: Witten pointed out that compact time dimensions occur in string theory; Hawking discussed the implications of closed time for cosmology; Bars developed two-time physics. And the rolled-up view touches thermodynamic time: If time is compact, the arrow direction is not fundamental, but an emergent property of the non-compact spatial parts — a natural explanation of macroscopic irreversibility without assumptions about initial conditions.

***For all readers:** One can build a clock in two ways. Either one hangs a pendulum on a frame and lets time run from outside — then one must adjust the length of the pendulum by hand. Or one builds the clock as a closed circle in which each oscillation returns into itself — then the circle itself determines which oscillations are possible. The FFGFT builds the clock in the second way. Therefore it does not need to adjust the masses: They fall out of the geometry.*

15.4 The Core of the Trade-off

What the unrolled view gains: the seamless connection to the machinery of quantum field theory; the freedom from a strong ontological commitment; the familiarity of the open time axis. **What it costs:** The structure alone does not fix any masses. Dynamical values must come from outside; predictions depend on external calibration; a model that understands itself as independent couples for its numbers back to measurement.

What the rolled-up view gains: The modes and their ratios fall out of the structure, without parametric attachment; the discrete quantization is a gift rather than a postulate. **What it costs:** a strong ontological claim — time as a closed dimension — that contradicts the accustomed intuition of running time. And it does not eliminate every external connection: The absolute scale still needs a unit bridge, the SI conversion via fixed conversion factors.

But the decisive finding is this: **Both paths use the same operator formalism.** The difference lies not in the mathematics, but solely in whether time sits inside or outside. One can keep the same construction — self-adjoint operator, spectral sectors — and only pull time into the space; then the sectors become modes, and the mass that previously had to be attached becomes an eigenvalue of the structure. The choice of time placement is a framework choice that precedes model building and determines its interpretive reach.

15.5 Who Has Already Pulled Time Inside

The idea of not treating time as an external parameter is not a novelty, but a long-followed line. It is honest to locate the rolled-up view in this line and to delimit its peculiarity.

Relational time. Page and Wootters let time arise within the state space: as entanglement between a clock subsystem and the rest; the total state is stationary, “evolution without evolution”. Moreva et al. and Marletto and Vedral have refined and experimentally illustrated the mechanism; Gambini and Pullin developed the Montevideo interpretation.

Timeless quantum gravity. Wheeler–DeWitt: The universal wave function is timeless, $\hat{H}\Psi = 0$; the resulting “problem of time” is the statement that there is no external time. Kuchař and Isham provided the overviews. Rovelli: relational quantum mechanics, with Connes the thermal time hypothesis. Barbour: the consistent elimination of time in favor of timeless configurations.

Compact time in string theory. Bars and colleagues with two-time physics and extended symmetry group.

The mode side. Wigner: Mass is a Casimir invariant, a label of irreducible representations of the Poincaré group — already a structural eigenvalue. Kaluza and Klein: A compactified dimension

generates a discrete mode tower with quantized charges — the blueprint for the fact that a rolled-up dimension brings forth modes.

The demarcation is precise: The majority of this tradition makes time relational, emergent, or timeless — it takes time as fundamental *out*. The rolled-up view takes time *in*: as a compact geometric dimension that carries modes. It shares with Page–Wootters and Rovelli the rejection of the external parameter time, but differs in the positive proposal.

15.6 Mathematical Execution

The Hilbert space of the rolled-up view is $\mathcal{H} = L^2(T^1 \times M) \otimes \mathcal{H}_{\text{int}}$, with a compact time dimension T^1 of length T and the ordinary space M . A function on it expands in a Fourier series:

$$\Psi(t, \mathbf{x}) = \sum_{n \in \mathbb{Z}} e^{int/T} \psi_n(\mathbf{x}). \quad (15.1)$$

The discrete frequencies $\omega_n = 2\pi n/T$ correspond to the mass values in natural units. The operator that generates them is $\hat{E} = -i \partial_t$ with the spectrum $\{2\pi n/T\}$. The quantization of masses is a direct consequence of the periodicity of time; nothing needs to be postulated.

Yukawa coupling as selection rule. The coupling between modes is a matrix element of the interaction operator:

$$g_{ijk} \propto \int_{T^1} e^{i(n_i - n_j - n_k)t/T} dt = T \cdot \delta_{n_i, n_j + n_k}. \quad (15.2)$$

This is an exact conservation of the mode index in the time direction. The Yukawa couplings of the Standard Model appear as selection rules for the mode indices, not as free parameters. [B]

Charge quantization. The compact time direction behaves like a fifth dimension in the sense of Kaluza–Klein, but timelike. The charge is $Q \propto n/R_T$ with the radius R_T of the compact time — charge quantization as a consequence of periodicity.

Mass hierarchy. The corpus describes the lepton masses in three mutually translatable representations: first, fundamentally via quantum numbers (n_i, l_i, j_i) with $\xi_i = \xi_0 f(n_i, l_i, j_i)$ and $E_i = 1/\xi_i$, where the absolute values carry the fractal correction K_{frac} ; second, in ladder form $m_i = r_i \xi^{\pi_i} \nu$ with rational r_i and the generation hierarchy $\pi = \frac{3}{2}, 1, \frac{2}{3}$; third, as $\mathbb{Z}_3$ circulant, $\sqrt{m_k} = M(1 + \sqrt{2} \cos(\frac{2}{9} + 2\pi k/3))$, with

the Koide relation $Q = 2/3$ and the angle $\theta = 2/9$ as output of the diagonalization. The dimensionless ratios are exact; deviations arise only in the transition to absolute SI values, from the SI conversion, the bridge constants ν and E_0 , the fractal correction, and the finite measurement precision. That is a structural proof: The ratios follow from the geometry, the absolute values from the bridge. [K]

15.7 Critical Appraisal

Ontological status. The compact time is real and geometric in the corpus, not merely computational technique. The boundary $\partial T^4 = \emptyset$ makes the question of an initial time point not wrongly answered, but wrongly posed. Time is discrete: The FFGFT possesses a minimal, by ξ fixed time increment $d_1 = 100\xi_0 = 1/75$, and the admissible states are discrete modes. Wheeler's "time as illusion" is not shared by the rolled-up view: It takes time as real, but geometrically different than accustomed.

The absolute scale: fixed, not free. The mass ratios follow from the mode structure and are exact. The absolute scale is not a free parameter: The ratio framework rests on $\xi_0 = 4/30000$, the absolute values carry their SI correction via fixed bridge constants — ν in the masses, $E_0 = 7,398 \text{ MeV}$ in $\alpha = \xi E_0^2$, fixed numerical factors in G . The Planck quantities are not an input, but follow from ξ : $\xi \rightarrow \{m_e, \dots\} \rightarrow G \rightarrow \ell_P$, and $\hbar$ itself is geometric. The only exact ξ power relation is $L_0/\ell_P = \xi_0$, $L_0 \approx 2,15 \times 10^{-39} \text{ m}$. The time cycle is not coupled to the Planck scale — the causal direction is reversed. No modulus remains open to be calibrated; the dilaton analogy to string theory does not carry here.

Causality. Does a periodic time generate closed timelike curves? The carrier is a torus with $\pi_1 \neq 0$ and stable windings, not a simple circle; the relevant connection is the topological connection on T^4 (Chapter 3). The classical CTC problem of the Gödel solutions concerns the decompactified Lorentz sector; the compact mass-time direction is spectral, not causally closed.

Experimental verification. The rolled-up view predicts no hidden heavy mode tower above the accelerator reach. The discrete spectrum of compact time *is* already the observed particle spectrum via $\tilde{T} \cdot m = 1$; there is no additional Kaluza-Klein tower. Testable

are: the particle masses themselves as ξ derivations (lepton structure, Koide at 10^{-5} , $g - 2$); the mode index conservation as coupling selection rule; the CHSH amplitude of order ξ ; the fractal dimension D_f and the fundamental length L_0 . For a clean falsification, ratio tests alone are not sufficient — only a forward-fixed, independently testable prediction is robust.

15.8 The Hidden Inconsistency of Standard QM

Instantaneity is not a QM-specific problem, but an inheritance of classical physics. In Newtonian mechanics, gravitation acts instantaneously; the Coulomb potential is instantaneous; Lagrange and Hamilton mechanics describe the state $(q(t), p(t))$ at a fixed t — the calculation is a timeless algebraic evaluation. No one asks how long this calculation takes, because calculations in this language have no duration. Pauli only made explicit what was classically silent.

QM inherits this and adds a tension. **The state $\psi(t)$ is instantaneous:** a timeless quantity at a fixed t , no duration, no process. **The evaluation $|\psi(t)|^2$ takes time:** To verify a probability density, one needs many measurements, thus real physical time. The formalism is instantaneous, its verification is temporally extended. The time that is needed for the probabilistic evaluation stands outside the formalism — it is neither operator nor variable, but what the experimenter expends to even know the state.

Superposition $c_1|\psi_1\rangle + c_2|\psi_2\rangle$ is thus doubly timeless: as a state (no time operator, no intrinsic “when”) and as a statement (probabilistic, needs time for verification, which lies outside). What exists as superimposed can only be evidenced through repeated measurements over real time — and every measurement destroys the superposition.

15.9 Three Consequences That Are Rarely Stated

In textbooks, time is introduced as a parameter without emphasizing what the absence of a time operator means. It means concretely:

Superposition is timeless. An electron in the state $c_\uparrow|\uparrow\rangle + c_\downarrow|\downarrow\rangle$ has no intrinsic time scale. The question “how long is it in this superposition?” is not poseable in the formalism. What exists is the coherence time T_2 — a phenomenological quantity of the environment coupling, not an intrinsic property of the state. In the FFGFT, each mode carries $\tilde{T}_i = 1/m_i$ as a geometric quantity. A state *is* its time scale — not as the duration of a superposition, but as a structural property of the mode.

Entanglement without simultaneity. Two entangled particles share a state in $\mathcal{H}_1 \otimes \mathcal{H}_2$ without built-in time structure. The question of the simultaneity of a measurement on both is not a QM question, but a classical one about the external parameter — and thus a relativistic one. That is the deepest reason why QM is compatible with Lorentz invariance: Simultaneity stands outside the formalism. In the FFGFT: If two modes with $m_1 \neq m_2$ were entangled, they would have different $\tilde{T}_1 \neq \tilde{T}_2$. Cross-mode entanglement generates incompatible time scales — structurally more problematic than in QM, where the conflict does not arise because t stands outside.

Collapse without duration. The transition $|\psi\rangle \rightarrow |\psi_k\rangle$ has no duration in the formalism; it is instantaneous in t . How long the collapse takes lies outside standard QM. Without a time operator there is no observable “collapse duration”. In the FFGFT, the Type I step (measurement act) is a geometric transition: The mass circle S_m^1 becomes the time axis $\mathbb{R}_t$, the winding number is discarded. This is an information loss with a natural direction.

15.10 What $\tilde{T} \cdot m = 1$ Structurally Changes

The fundamental relation sets time not as an external parameter, but as a geometric quantity of each mode: The period $R_m = \hbar/(mc^2)$ is intrinsic — it belongs to the particle like its mass. Three consequences:

Superposition within a mode — intermediate torsion — has a natural time scale $\tilde{T}_i$. Superposition *over* different modes would superimpose incompatible time scales; that is the structural reason why cross-mode superpositions are problematic in the FFGFT. [B]

Entanglement within a mode has simultaneity defined by $\tilde{T}$. Across modes, a common time concept is lacking — similar to

QM, but for a different reason: not because t stands externally, but because different modes carry different geometric time scales.

Collapse (Type I) has a natural direction — lossy from S_m^1 to $\mathbb{R}_t$ — and a natural asymmetry: The reverse direction requires the discarded winding number. The time arrow is not an additional assumption, but a consequence of the geometry of measurement.

The Pauli theorem applies to an open time parameter. For a compact time dimension it does not apply: Its conjugate has a discrete spectrum, the winding numbers. $\tilde{T} \cdot m = 1$ realizes exactly this situation: Time is compact, the conjugate spectrum is the discrete mass ladder. The theorem does not exclude this case — it enforces it as the only consistent way to integrate time into the Hilbert space.

15.11 Unrolling: Type I and Type III

Two different operations carry the same name. **Type I unrolling** is the measurement act: lossy, the projection from the compact carrier onto the real axis is not reversible. That is the collapse. **Type III unrolling** is the representation change: the bijective pullback from Chapter 13, no information loss, only a language change. The correction factor K_{frak} sits in the mass formula of the Type I transition; it is not involved in the Type III pullback. Anyone who calculates in Hilbert space language sees K_{frak} only in the SI translation. [B]

15.12 Summary

	Standard QM	FFGFT
Time	absent: no operator; external parameter	geometric: $\tilde{T}_i = 1/m_i$, compact, mode- bound
Superposi- tion	timeless, no intrinsic clock	intermediate torsion in a mode; $\tilde{T}_i$ as scale
Cross-mode	algebraically allowed, no time conflict	incompatible $\tilde{T}_i \neq \tilde{T}_j$, structurally problem- atic
Entangle- ment	timeless; simultane- ity external (Lorentz)	mode-specific $\tilde{T}_i$; no common time concept across modes
Collapse	instantaneous in t , no duration	Type I: directed infor- mation loss, time ar- row geometric
Pauli theo- rem	blocks time operator for open t	does not apply for compact $\tilde{T}$

The choice of where time sits precedes the mathematics and decides whether the modes are given or attached. Anyone who reads the sectors as spatial eigenspaces keeps the familiar physics and pays with the external attachment of the numbers. Anyone who takes time as a closed dimension into the space obtains the modes structurally and pays with an ontological commitment. The ontological price of the second choice remains on the table — but the methodological price of the first is not smaller, it is only more familiar.

Sources: Doc. 307 (Time in state space, July 2026), Doc. 334 (Superposition without time, Aug. 2026), Doc. 333 (Type I/III), Doc. 306 ($T \cdot E = 1$), Doc. 270 (Decompactification)

Symbols and Abbreviations

Basic Quantities of the FFGT

ξ	Geometry parameter of the FFGT, $\xi = 4/30000 = 1/7500 \approx 1,333 \times 10^{-4}$. The only free parameter; follows from Galois group orders of the carrier.
$\tilde{T}$	Compact time coordinate of a mode; $\tilde{T} = 1/m$ in natural units ($\hbar = c = 1$).
$T^4/\mathbb{Z}_3$	FFGT carrier: four-dimensional torus with $\mathbb{Z}_3$ symmetry. Three spatial and one temporal winding direction; nine fixed points, triality symmetry.
K_{frak}	Fractal correction factor, $K_{\text{frak}} = 1 - 100\xi \approx 0,9867$. Arises during the Type I transition from the compact carrier to the SI scale; cumulative over 17 orders of magnitude (RG flow from Planck to electroweak scale).
D_f	Fractal spatial dimension, $D_f = 3 - \xi \approx 2,9999$. Local; not to be confused with K_{frak} .
L_0	Fundamental FFGT length, $L_0 = \xi \ell_P \approx 2,15 \times 10^{-39}$ m.
ℓ_P	Planck length, $\ell_P \approx 1,616 \times 10^{-35}$ m.
t_P	Planck time, $t_P \approx 5,39 \times 10^{-44}$ s.

Qubit and Quantum Computer Symbols

- (z, r, θ) Cylindrical phase space coordinates of an FFGFT qubit: $z \in [-1, +1]$ projection onto the computation axis, $r \in [0, 1]$ radial superposition amplitude, $\theta \in [0, 2\pi)$ relative phase.
- (α, β) Standard QM amplitudes of a qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$.
- K_{frak} **(gate)** In the π gates (σ_x rotation): rotation angle $\alpha = \pi K_{\text{frak}}$ instead of π . Derived geometrically from D_{fi} ; testable prediction.
- $\hat{F}$ Fundamental fractal spectral operator on $\mathcal{H} = L^2(T^4/\mathbb{Z}_3) \otimes \mathbb{C}^3$. Self-adjoint; $\xi = \lambda_{\min}(\hat{F}_{D_4})$.
- D_4 Root lattice in $\mathbb{R}^4$: all integer vectors with even coordinate sum. Kissing number 24; basis of the mode spectrum.
- CHSH Clauser-Horne-Shimony-Holt parameter; local-realistic ≤ 2 , Tsirelson bound (QM) $2\sqrt{2} \approx 2,828$.
- $\Delta\text{CHSH}_{\text{elem}}$ Elementary FFGFT deviation per measurement: $\xi/(2\pi) \approx 2 \times 10^{-5}$. Hypothesis; formal derivation pending.
- T_1, T_2 Relaxation time (T_1) and coherence time (T_2) of a qubit; always $T_2 \leq 2T_1$.
- Q, Q_ϕ Register dimension: $Q = 2^n$ (standard QFT), $Q_\phi = \phi^n$ (ϕ -QFT). Both equivalent for period finding up to $N < 2^{20}$ on $\pm\xi$.

Spectral Theory and Lattices

- $N(n)$ Degeneracy of the shell $|k|^2 = n$ in the D_4 lattice; coefficient of the theta series $\theta_{D_4} = (\theta_3^4 + \theta_4^4)/2$.
- $Z_{\mathbb{Z}^4}(s)$ Spectral zeta of the cubic torus: $8(1 - 4^{1-s})\zeta(s)\zeta(s-1)$.
- $\text{Aut}(D_4)$ Automorphism group of the D_4 lattice; $|\text{Aut}(D_4)| = 1152 = 2^7 \cdot 3^2$. Coprime to 5: no fivefold symmetry.

- χ_0, χ_1, χ_2 The three $\mathbb{Z}_3$ characters of the orbifold; each D_4 shell decomposes uniformly into $N(n)/3$ states per sector.
- λ_4 Arc length on the mass circle; de Broglie relation $\lambda_4 \cdot m = 2\pi$ exact per mode (position relation, not time relation).

Time Structure

- Type I** Measurement act (unrolling): lossy, $S_m^1 \rightarrow \mathbb{R}_t$. Information loss, time arrow geometric.
- Type III** Representation change (pullback): lossless, bijective. No information loss; K_{frak} not involved.
- ξ_{Higgs} Higgs coupling parameter, $\approx 1,038 \times 10^{-5}$. Acts in the algebraic state space (coupling rates, decoherence); *not* identical with ξ , which acts in geometric position space.

Status Markers

- [K]** Confirmed — fully proven from ξ or a corpus-secured derivation; test script passed.
- [B]** Documented — numerically secured; analytical proof present or trivial.
- [S]** Speculative — hypothesis or indication; not yet fully derived.
- [E]** Established physics — standard result, here only recalled.
- [X]** Checked negative — explicitly excluded with justification.
- [SETZUNG]** Axiomatic postulate — starting point of the derivation, not further justified.

Bibliography

FFGFT Corpus Documents

All documents are freely accessible in the repository.

PDFs (corpus documents):

<https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/pdf/>

This book (De/En, LaTeX sources):

<https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/Sources/qmb-ch/>

Verification scripts (Python):

https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/python/QMB_Skripte/

The following list enumerates the corpus documents used in this book in numerical order.

Doc. 022 T0 QFT-ML-Addendum: Cumulative CHSH formula (2025)

Doc. 023 Bell tests and ξ -modified correlations (Oct. 2025)

Doc. 023a/b Bell tests Part 2; Response to the Bell video (2025)

Doc. 034 Geometric qubit formalism (Nov. 2025)

Doc. 055 Dynamic mass, photons, nonlocality (2025)

Doc. 073 Four algorithms in the FFGFT framework (2025)

Doc. 074 No-go theorems and FFGFT (2025)

Doc. 075/076 RSA method; RSA period tests (2026)

Doc. 083 Photon chip classification (2025)

- Doc. 131** Apparent instantaneity — field dynamics instead of action at a distance (2026)
- Doc. 133** Fractal correction K_{frak} : Derivation and RG flow (2026)
- Doc. 147** Quantum computer: ϕ -QFT and IBM Heron r2 (May 2026)
- Doc. 161** Coherent Ising Machine and FFGFT (2026)
- Doc. 173** Quantum dot as resonator (March 2026)
- Doc. 174** Spin and qubit states (2026)
- Doc. 175** Qubit state spaces and photonic hierarchy (March 2026)
- Doc. 176** Shor's algorithm: Preparation vs. transformation (2026)
- Doc. 178** Spin stability and permanent magnet (2026)
- Doc. 183** Google Willow: topological error correction (April 2026)
- Doc. 184** p-bit: between bit and qubit (2026)
- Doc. 190** Correction register — Precisions 11 and 12 (May 2026)
- Doc. 230** Translatability FFGFT $\leftrightarrow$ Hilbert space (May 2026)
- Doc. 231** Hilbert space extension (2026)
- Doc. 285** Dimension bridge HLV (2026)
- Doc. 291** Dynamic location $\theta = 2/9$ (2026)
- Doc. 293** Icosahedron and phase $\theta = 2/9$ (2026)
- Doc. 306** Native time-energy reciprocity $T \cdot E = 1$ (2026)
- Doc. 307** Time in state space: rolled up vs. unrolled (July 2026)
- Doc. 311** Four onto three: $24 = 12 + 12$ (2026)
- Doc. 313** No beginning: Closure bifurcation (2026)
- Doc. 314** The lattice in Hilbert space (2026)
- Doc. 316** Weyl law and Riemann hypothesis (Aug. 2026)
- Doc. 322** Spectral theory on $T^4/\mathbb{Z}_3$ (2026)
- Doc. 327** Self-adjointness of $\hat{F}$ (Aug. 2026)
- Doc. 333** K_{frak} , unrolling and $\tilde{T} \cdot m$ duality (Aug. 2026)
- Doc. 334** Superposition without time (Aug. 2026)

Doc. 338 Mass calculations and Galois comparison (2026)

Doc. 339 Frobenius separation massive/massless in $GF(27)^*$ (Aug. 2026)

Doc. 343 Zeta function, resolution limit, quantum computer (Sept. 2026)

Doc. 353 Koide amplitude $d/c = \sqrt{2}$ from $T^4/\mathbb{Z}_3$ geometry (Sept. 2026)

Doc. 364 Icosahedron/D4 comparison, Galois structure (Sept. 2026)

Doc. 365 Foundation of the FFGFT (Sept. 2026)

Selected External Sources

Bell 1964 J. S. Bell: *On the Einstein-Podolsky-Rosen paradox*. Physics 1 (3), 195–200 (1964). — The original argument for the impossibility of local-realistic hidden variables.

CHSH 1969 J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt: *Proposed experiment to test local hidden-variable theories*. Phys. Rev. Lett. 23, 880 (1969). — Experimentally testable form of the Bell inequality.

Aspect 1982 A. Aspect, P. Grangier, G. Roger: *Experimental realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment*. Phys. Rev. Lett. 49, 91 (1982). — First loophole-free Bell test (loopholes fully closed in 2015).

Kochen/Specker 1967 S. Kochen, E. P. Specker: *The problem of hidden variables in quantum mechanics*. J. Math. Mech. 17, 59–87 (1967). — Proof of the context-dependence of observables.

PBR 2012 M. F. Pusey, J. Barrett, T. Rudolph: *On the reality of the quantum state*. Nature Physics 8, 475–478 (2012). — Shows: the quantum state is ontic, not a state of knowledge.

Shor 1994 P. W. Shor: *Algorithms for quantum computation: discrete logarithms and factoring*. Proc. 35th FOCS, 124–134 (1994). — Polynomial-time factorization on a quantum computer.

Grover 1996 L. K. Grover: *A fast quantum mechanical algorithm for database search*. Proc. 28th STOC, 212–219 (1996). — $O(\sqrt{N})$ search in an unsorted database.

Loss/DiVincenzo 1998 D. Loss, D. P. DiVincenzo: *Quantum computation with quantum dots*. Phys. Rev. A 57, 120 (1998). — Spin-qubit gates via exchange interaction.

- Willow 2024** Google Quantum AI: *Quantum error correction below the surface code threshold*. Nature 634, 2024. — First surpassing of the error correction threshold on Willow (105 qubits, IBM Heron comparison).
- Will 2025** Will et al.: *Floquet topological order in a quantum processor*. Nature, Sept. 2025. — Non-equilibrium topological order (chiral edge modes, anyons) on Willow.
- Solovay/Kitaev** A. Y. Kitaev: *Quantum computations: algorithms and error correction*. Russ. Math. Surv. 52 (1997). — Universality of $\{H, T, \text{CNOT}\}$ for quantum computing.
- Tsirelson 1980** B. S. Tsirelson: *Quantum generalizations of Bell's inequality*. Lett. Math. Phys. 4, 93–100 (1980). — Upper bound of quantum correlations: $2\sqrt{2}$.
- Kaluza 1921 / Klein 1926** T. Kaluza (1921), O. Klein (1926): Compactified dimension generates discrete mode tower and charge quantization. Mathematical blueprint for the time-mass mechanism of the FFGFT.
- Page/Wootters 1983** D. N. Page, W. K. Wootters: *Evolution without evolution*. Phys. Rev. D 27, 2885 (1983). — Time as entanglement correlation within the state space.
- Weyl 1912** H. Weyl: Eigenvalue law of compact manifolds $N(\lambda) \sim C\lambda^{d/2}$. Establishes the impossibility of carrying Riemann zeros as a Laplace spectrum.
- Inagaki 2016 / McMahon 2016** T. Inagaki et al.; P. L. McMahon et al.: Coherent Ising Machine (CIM) with DOPO network. Science 2016. — Experimental realization of physical optimization through relaxation.
- Schiemann 1997** A. Schiemann: Isospectral, non-isometric tori in dimension 4. Shows that the spectrum alone does not determine the lattice. Limit of the informative value of spectral methods.
- BSI 2023** Federal Office for Information Security: *Status of quantum computer development*. 2023. — Current assessment of the threat to cryptographic methods.

Note on the Origin

This book grew out of the FFGFT corpus, which has comprised over 365 individual documents since 2025. The texts of the individual chapters were conceived and substantively overseen by Johann Pascher; Claude (Anthropic, Sonnet 4.6) assisted with the formulation, with the merging of the source sections, and with the compilation of the $\text{\LaTeX}$ template.

What AI assistance meant in this project: The source documents (Doc. 022 to Doc. 365) contain the actual derivations, test scripts, and epistemic markers. The AI read these documents, reformulated the chapters in its own language — without verbatim copying, without external content — and checked the consistency of the status markers. All substantive decisions, boundary definitions, and status markers are Johann’s work; the AI was a formulation tool, not a substantive author.

What AI assistance did not mean: None of the physical statements originate from the AI’s training corpus. All numbers, derivations, and cross-references trace back to the aforementioned source documents, which are publicly viewable in the repository. Questions as to whether a result is “new” or “correct” were answered by the author himself, not by the AI.

Verifiability: Each chapter bears at the end a quelle reference to the underlying corpus documents. Each substantive statement bears a status marker ([K], [B], [S], [E], [X]). Anyone who wishes to question a statement will find the complete derivation path with test script in the cited document.